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The Trouble with Instruments: The Need for Pretreatment Balance in Shock-Based Instrumental Variable Designs

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Abstract. Credible causal inference in accounting and finance research often comes from natural experiments. These experiments can be exploited using several shock-based research designs, including difference in differences (DID), shock-based instrumental variable (shock-IV), and regression discontinuity. We study here shock-IV designs using panel data. We identify all shock-IV papers in two broad data sets and reexamine three of the apparently *strongest* papers—Desai and Dharmapala [Desai M, Dharmapala D (2009) Corporate tax avoidance and firm value. *Rev. Econom. Statist.* 91:537–546.], Duchin et al. [Duchin R, Matsusaka J, Ozbas O (2010) When are outside directors effective? *J. Financial Econom.* 95:195–214.], and Iliev [Iliev P (2010) The effect of SOX Section 404: Costs, earnings quality, and stock prices. *J. Finance* 65:1163–1196.]. After we enforce covariate balance and common support for treated and control firms, the instruments in all three papers are unusable—they are no longer significant in the first stage. All three papers also show nonparallel pretreatment trends on outcomes or core covariates. The problems with these papers generalize to our full sample and to other papers exploiting the same shocks as Duchin et al. A core conclusion of our reexamination is that pretreatment balance (common support, covariate balance, and parallel pretreatment trends) is necessary for credible shock-IV designs. We provide a good-practice checklist for shock-IV design with panel data, much of which also applies to DID designs.

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Keywords: instrumental variables • shock-based research design • covariate balance • common support • parallel trends

1. Introduction

Increasingly, accounting and finance scholarship stresses the identification of causal effects, often through studies that exploit shocks that plausibly satisfy the core conditions for credible causal inference (Atanasov and Black 2016, Bowen et al. 2017). However, current practice in exploiting shocks often neglects the need for *pretreatment balance* between treated and control firms—a concept that we define as common support, balance on pretreatment covariates and outcomes, and parallel pretreatment trends.

In this paper, we study shock-based instrumental variable (shock-IV) designs using panel data, where the instrument relies on an underlying shock. To assess practice when using shock-IVs, we began by searching for shock-IV papers in major finance and accounting journals over 2001–2015 (see the online appendix for details). This search generated 32 shock-IV papers, of which 26 had panel data. We chose for close reexamination, after an initial read, three of the apparently

strongest papers from this sample. These were papers that we liked from authors who we know and respect: Duchin, Matsusaka, and Ozbas (DMO; Duchin et al. 2010); Iliev (Iliev 2010); and Desai and Dharmapala (D&D; Desai and Dharmapala 2009). All three use plausible instruments, address important issues, and are careful in many ways. The authors were generous enough to share their data sets and code (and went to considerable trouble to do so).¹

We found that after we enforce common support and balance on pretreatment covariates and outcomes, the IVs in all three papers (and a family of other papers using the same IVs as DMO) become too weak to be usable. All three papers also exhibit nonparallel pretreatment trends on outcomes or core covariates. We use our reexamination of these papers to explore what can go wrong if pretreatment balance is absent and develop a checklist for shock-IV designs that stresses the need for pretreatment balance. Many checklist items also apply to difference-in-differences (DiD) and non-shock-IV designs.

DMO study the effect of board independence on firm value and profitability. Their principal instrument for a change in board independence is whether a firm had a less than 100% independent audit committee in 2000. Firms had to adopt fully independent audit committees to meet New York Stock Exchange (NYSE) and National Association of Securities Dealers Automated Quotations System (NASDAQ) requirements adopted in 1999 (“Audit Committee Shock” and related “*Audit Committee IV*”). DMO hypothesize that this caused firms to add independent directors to their boards to staff the audit committees. They find that a higher percentage of independent directors (*PctIndep*) is value neutral overall but positive (negative) for firms with low (high) values for a measure of information costs (*InfoCost*). DMO find similar results with a “*Minority-Independent IV*” based on stock exchange rules adopted after the Sarbanes–Oxley Act of 2002 (SOX), which required public firms to have a majority of independent directors (“SOX Board Shock”), and in their response to a prior draft of this paper (Duchin et al. 2015), a “*Distance-to-Compliance IV*.”² We term these three IVs the *Board Independence IVs*. A family of other papers exploits these shocks using shock-IV (with one or more of these instruments) or DiD designs.³

The Board Independence IVs have an array of problems. We consider the *Audit Committee IV* in this introduction and the others in the text and the online appendix. The DMO sample has limited common support: almost no control firms (with fully independent audit committees) have values for *PctIndep* < 25%, which DMO measured in 2000 (*PctIndep*₂₀₀₀), whereas almost no treated firms (without fully independent audit committees preshock) have *PctIndep*₂₀₀₀ > 80%. The DMO sample also has severe imbalance between treated firms and control firms on *PctIndep*₂₀₀₀: treated firms had far fewer independent directors than control firms. These imbalances drive their results. If we impose common support and then compare treated and control firms with similar *PctIndep*₂₀₀₀, *Audit Committee IV* loses first-stage strength—it no longer predicts a significant change in their instrumented variable, which is change board independence from 2000 to 2005 ($\delta Indep$).

Yet, if we do not require common support and covariate balance on *PctIndep*₂₀₀₀—the approach that Duchin et al. (2015) advocate—*Audit Committee IV* has severe problems. Its apparent power is spurious—driven by regression of *PctIndep* to a time-varying mean. *Audit Committee IV* also fails placebo tests: it predicts the instrumented variable $\delta Indep$ (1) during the preshock period (when no shock existed) and (2) looking *backward in time* from the shock to earlier years. Pretreatment trends are also strongly nonparallel. The deficiencies of the Board Independence IVs have implications for other board composition

shocks, such as rules requiring a minimum percentage of women on company boards⁴ and, more generally, any IV defined based on a regulatory threshold that firms might tend to approach voluntarily.

Iliev (2010) studies the cost of compliance with SOX section 404 for firms near the compliance threshold (public float of \$75 million (M)) using a combined regression discontinuity (RD)–IV design. His RD design exploits the discontinuity at \$75M in float. Iliev finds that some firms manipulate their float to stay below the \$75M threshold and uses a clever IV to address manipulation: whether a firm had float >\$75M in 2002. This instrument exploits a Securities and Exchange Commission (SEC) rule adopted in 2003, which required SOX section 404 compliance by firms with float >\$75M in 2002 (before the \$75M rule was known).⁵

Iliev’s core RD design is creative. However, his IV generates strong imbalance between IV-complier and control firms on pretreatment growth. If we require common support on growth, the sample is too small for IV estimation. Instead, we apply a principal strata approach (Frangakis and Rubin 2002) using strata based on growth over 2002–2004 and conduct a combined DiD/RD analysis within strata. We estimate that SOX section 404 compliance increases audit fees for firms near the \$75M threshold by around 80% versus Iliev’s preferred RD/IV estimate of a 167% increase; his higher estimate is driven by imbalance on growth.

D&D study how corporate governance mediates the effect of sheltering income from tax on firm value. Their shock is 1996 U.S. Treasury tax rules directed at small private firms, which increased tax shelter opportunities for multinational firms. D&D interact this shock with measures of the firm’s need to shelter income to create instruments for *book–tax gap* (BTG; their proxy for tax sheltering activity). They find that increased sheltering opportunities increase firm value but only for firms with high institutional ownership.

We apply the D&D IVs to a pre–post balanced sample of firms that appear in their data set both before and after the shock. For this sample, their IVs, already weak, no longer significantly predict the instrumented variable (book–tax gap) in the first stage. Their instruments are also not exogenous (despite an exogenous shock), and their sample exhibits covariate imbalance and nonparallel pretreatment trends.

A number of themes emerge from our reanalysis. We summarize these in a checklist for shock-IV design with panel data. These include (1) the crucial role of pretreatment balance for shock-based designs using panel data, including IV and DiD, (2) the value (rooted in causal inference theory) of trimming to common support and balancing on relevant preshock covariates and outcomes, (3) the value of using both DiD and shock-IV in the same study when feasible, (4) the importance of falsification tests, (5) for a shock

based on a threshold, the value of using RD methods and assessing whether regression to the mean (drift relative to the threshold) can explain apparent shock power, (6) if the shock is interacted with a covariate, the need to confirm parallel trends for the double differences as well as the third difference, and (7) when the shock and the outcome are separated in time, the time gap can introduce imbalance, even if the sample is balanced at the time of the shock.

The themes from our three reexaminations generalize to the other panel data papers in our sample. None of these papers verify common support. Only five papers (DMO, Iliev, and three others) check for either covariate balance or parallel pretreatment trends; only DMO and Tsoutsoura (2015) check both. Similarly, only a few papers carefully defend shock exogeneity (D&D, Iliev, and six others) or the only-through condition (D&D, Iliev, and four others). DMO, Iliev, and D&D are indeed among the strongest shock-IV papers in our sample.

We build here on the Atanasov and Black (2016) survey of shock-based research designs. Larcker and Rusticus (2010), Roberts and Whited (2013), Gow et al. (2016), and Jiang (2017) discuss challenges in finding valid IVs in finance and accounting research, but they do not focus on shock-based instruments. Young (2018) assesses IV validity for economics, and Sovey and Green (2011) do so for political science, but they also do not focus on shock-IVs. Bertomeu et al. (2016) note that shocks based on pretreatment compliance with a regulatory threshold are often not randomly assigned. Bakke and Whited (2012) discuss the need to limit bandwidth around a threshold in an RD design to improve balance. Angrist and Pischke (2015) stress the need for random assignment of IVs and see covariate balance as a check for random assignment.

We shared our analyses with D&D, DMO, and Iliev before circulating this article. All three responded privately; DMO also responded publicly (Duchin et al. 2015). DMO defend their choice not to address imbalance on *PctIndep*. Their defense is inadequate, as we discuss later.

This article focuses on pretreatment balance in shock-IV designs. The online appendix provides an expanded checklist for shock-IV design, discusses selected other papers in our sample, and presents further results from our reexamination of DMO, Iliev, and D&D.

2. Background on Shock-Based Research Designs and Shock-IV

We offer here a condensed review of shock-IV designs using panel data; see Atanasov and Black (2016) and the online appendix for additional details and literature references. We focus on firm-level analyses and assume

a binary shock: $w_{it} = 1$ in the postshock period if firm i is treated (subject to the shock) and 0 otherwise.

2.1. Conditions for Shock-IV Designs with Panel Data

Causal analyses typically seek to estimate the causal effect τ_i (for firm i) of treatment on an outcome y_{it} , where τ_i is the value of y_i if firm i is treated minus the value of y_i if firm i is not treated. With multiple time periods t ,

$$\tau_{it} = y_{it}(w_{it} = 1) - y_{it}(w_{it} = 0).$$

However, in each time period, we observe only one of the two potential outcomes, $y_{it}(w_{it} = 1)$ and $y_{it}(w_{it} = 0)$. In the postshock period, we must impute the missing potential outcomes using data from control firms. The central challenge to imputation is selection bias: treated and control firms may differ in ways, perhaps unobserved, that bias the estimated treatment effects. A key theme of this paper is that shock-IV and other shock-based designs provide credible causal estimates only if they approximate the conditions, including common support and covariate balance, that one would achieve from a true randomized experiment.

Atanasov and Black (2016) develop conditions for a good shock—one that approximates a randomized experiment. We refine and expand those conditions here for panel data designs.

1. **Shock strength.** The shock must be strong enough to significantly change firm behavior or incentives after imposing covariate balance and common support.

2. **Exogenous shock.** The shock came from outside the system that one is studying. Treated firms did not choose whether to be treated and could not change their behavior to anticipate the shock, the shock is expected to be permanent, and there is no reason to believe that potential outcomes or which firms were treated depend on unobserved firm characteristics.

3. **Only-through condition(s).** The effect of the shock on the outcome must come *only through* the shock. No other shock, at around the same time, should affect treated firms differently than control firms. If—as in an IV analysis—one expects the shock to affect outcomes through a particular instrumented variable, the shock must affect the outcome only through that variable. In IV analysis, this is often called an *exclusion restriction*; we prefer the term *only-through condition*.

4. **Pretreatment balance.** Exogeneity and the only-through condition will be credible only if treated and control firms are similar preshock, which we term *pretreatment balance*. This balance has several components.

a. *As if random assignment.* The shock should separate firms into treated and controls in a manner that appears to be close to random. One often needs to

allow an exception for the forcing variable that determines which firms are affected by the shock and closely related variables.

b. *Common support*. There should be common support—reasonable overlap between treated and control firms on relevant covariates and on the outcomes. Somewhat imperfect support can be addressed through sample trimming.

c. *Covariate balance*. The forcing and related variables aside, the shock should produce similar pretreatment values for treated and control firms on both relevant covariates and the outcome variable(s)—a concept that is often misleadingly called *covariate balance*. Somewhat imperfect initial balance can be addressed through balancing methods.

d. *Parallel pretreatment trends*. Treated and control firms should follow parallel trends on the outcome(s) and core covariates during the pretreatment period. For a triple-difference design, which interacts the shock with another variable, one should have parallel pretreatment trends for the relevant double differences as well as the third difference.

Conditions (1), (2), and (3) are well known for causal IV (Angrist and Pischke 2009, section 4.1).⁶ We focus here on condition (4)—the need for as if random assignment and its corollaries, covariate balance, common support, and parallel pretreatment trends. This need is less often discussed (for exceptions, see Rosenzweig and Wolpin (2000) and Angrist and Pischke (2015)).

2.2. IV Strength and Coefficient Blowup

Even if the only-through condition is fully satisfied, IV estimates can be biased, and standard errors are often downward biased if the IV or IVs are weak (Cruz and Moreira 2005). A common rule of thumb for IV strength is a first-stage *F*-statistic > 10 (Stock et al. 2002). Especially when the first stage is not strong, IV coefficients are prone to a *blowup* problem, with two-stage least squares (2SLS) coefficients much larger than ordinary least squares (OLS) coefficients from regressing the outcome on the instrumented variable (Jiang 2017). A high 2SLS/OLS coefficient ratio is a warning sign for instrument invalidity, though not proof; the higher the ratio, the stronger the warning.

2.3. Similarity Between DiD and Shock-IV

There is a close connection between DiD and shock-IV using the same shock. DiD has several advantages. It relies on weaker only-through assumptions. DiD estimates an *intent-to-treat* effect of a shock on all firms to which the shock applies (often called the *reduced form*). In contrast, shock-IV requires the researcher to specify a channel, and it assumes that the shock affects the outcome only through this channel and provides an estimate only for compliers—firms with

behavior that is changed by the shock. DiD also forces researchers to use a pre–post balanced sample and comes with useful technology, including checking for parallel pretreatment trends and falsification tests, such as assessing the effects of placebo shocks and whether the actual shock predicts placebo outcomes.

However, IV also has advantages. One must show a first stage—the shock must strongly influence the instrumented variable. In contrast, a DiD design leaves the first stage implicit and lets authors assume a first stage that may not exist (Litvak et al. 2016). The IV design, because it posits a specific channel, can make it easier to assess whether economic magnitudes are plausible. Thus, there is value in applying both designs to the same shock. If the shock is strong, both should have similar statistical strength.

The parallel-trends assumption is well known for DiD but less so for shock-IV research. To see why both designs rely on this assumption, it is useful to consider the similarity between them. If the instrumented variable is binary (or is made so by binning), DiD provides an intent-to-treat design, which estimates the average effect of the shock on all firms subject to the shock (assigned to treatment). In contrast, IV provides a *local average treatment effect* (LATE) for *compliers*—firms that *complied* with the assignment by changing the instrumented variable in the predicted direction. Without covariates, the 2SLS coefficient from an IV analysis can be expressed as a Wald estimate—the ratio of (the intent-to-treat DiD coefficient δ_{DiD}) to (the first-stage IV coefficient from regressing the instrumented variable x on the IV z)

$$\hat{\beta}_{2SLS} = \frac{\hat{\delta}_{DiD}}{\hat{\beta}_{1S}} = \frac{\text{effect of instrument (shock) on } y}{\text{effect of instrument (shock) on } x}. \quad (1)$$

If we add covariates, the 2SLS and Wald estimates will diverge slightly but should be similar for an as if randomly assigned instrument. For a strong instrument, statistical strength should be similar as well: $\hat{\beta}_{2SLS}$ and $\hat{\delta}_{DiD}$ can have very different coefficients and standard errors but should have similar *t*-statistics. If the instrument is not strong, $\hat{\beta}_{2SLS}$ will usually have a lower *t*-statistic than $\hat{\delta}_{DiD}$. Much as shock-IV directly exploiting the shock is similar to DiD, shock-IV where the shock is interacted with firm attributes, as in D&D and DMO, is similar to a triple-differences design.

2.4. Common Support

Common support and covariate balance are crucial but often overlooked aspects of both DiD and shock-IV designs. An ideal shock will divide firms into treated and controls at random and thus satisfy these conditions (unusual draws aside). If so, then estimates without covariates are unbiased. Adding covariates can

improve precision but should neither cause bias nor be needed to prevent omitted variable bias (OVB). Many real-world shocks, however, produce some lack of common support, imbalance on pretreatment covariates or outcomes, or both. The best responses to these problems are to (1) trim the sample to common support and (2) improve balance through matching or another balancing method. One can hope that trimming and balancing using observed covariates will also improve common support and balance on unobserved covariates.

The literature on pure observational studies stresses the need to limit the sample to observations within a region of “reasonably thick” common support—the area for which a reasonable number of treated and control firms have overlapping values—and avoid using regression to extrapolate beyond common support (Iacus et al. 2012; Imbens and Rubin 2015, chapter 14). For simplicity, assume that one is interested in the average treatment effect on the treated. The common support question is whether the covariate values for the control firms overlap with the range for treated firms. One can estimate each firm’s propensity to be treated based on pretreatment covariates and then require common support on the propensity score. One can also require common support for selected covariates by keeping only treated firms i that are within the convex hull of the controls—for each covariate x^j , $x_i^j \in [\min_{\text{controls}}(x^j), \max_{\text{controls}}(x^j)]$.

2.5. Covariate Balance

In classical IV, one often conditions on covariates by adding them in both stages of 2SLS. One assumes (often without strong justification) that the covariates x_{-1} (all predictor values other than the variable of interest) are exogenous ($\text{Cov}(x_{-1}, \epsilon) = 0$) and that, after including them, the IV will meet the requirements that $\text{Cov}(z, \eta) = 0$ in the first stage and $\text{Cov}(z, \epsilon) = 0$ in the second stage; where η and ϵ are the first- and second-stage regression errors. The challenge is to find a set of covariates x_{-1} that makes these untestable conditions plausible. This normally calls for including a relatively extensive set of covariates that covary with the instrument, the instrumented variable, or the outcome. Good practice is to choose these covariates in a “design stage” of the analysis, with outcomes hidden so that they cannot influence the choice of covariates (Rubin 2008, Rosenbaum 2009, Imbens and Rubin 2015).⁷ The (untestable) hope is that controlling for observed covariates also controls for unobserved covariates. The literature on pure observational studies stresses the importance of generating treated and control groups that are similar on a broad set of observable covariates. This literature advocates explicit balancing on covariates rather than relying on regression alone.⁸

Adding covariates or matching on them is less simple in a causal IV setting in two respects. First, with panel data, one should generally include only time-varying covariates that are not intermediate outcomes, which can be affected by treatment. For these covariates, it can make sense to balance on pretreatment values but omit the covariate from the 2SLS regression. Intermediate cases—whether an important covariate might be weakly affected by the treatment—call for judgment, informed by theory, on which risk is greater—OVB if the covariate is omitted or “conditioning on outcome” bias if it is included plus robustness checks with and without the covariate. Second, to allow for heterogeneous causal effects, we should estimate LATE conditional on covariates (known as a *local average response function*) and then integrate over the covariate space to obtain an overall LATE estimate (Abadie 2003).⁹

Still, a core lesson, from both classical regression theory and observational study design, is the value of bringing an imperfect shock closer to being as good as randomly assigned by balancing on covariates that plausibly predict treatment or including them in a 2SLS regression. We discuss in Section 6 the exceptions, where one should not balance on an observed covariate that plausibly predicts treatment or should not include a covariate in a 2SLS regression.

2.6. Implications of Trimming and Balancing for Sample Size and Precision

Sample trimming has two principal costs. First, standard errors will increase because of reduced sample size. For example, trimming the DMO sample to common support (see Section 3) reduces the sample from 323 treated firms and 582 control firms to 267 treated firms and 343 control firms.¹⁰ However, the smaller standard errors that one can obtain with a larger sample using regression methods that extrapolate beyond common support are not appropriate measures of the confidence that one should have in one’s results. Second, coefficient estimates apply only to the trimmed sample. However, treatment effect estimates based on firms that are outside common support are not reliable in any case.

Balancing methods will also often reduce sample size. For example, the approach that we implement for DMO in Section 3 (one-to-one matching on the propensity score with replacement) leaves 267 treated firms and 267 matched controls (156 unique control firms) when it would potentially be possible to use all 343 control firms that remain after trimming. However, here too, unless balance (after trimming) is already excellent, there is usually no good alternative. A treatment effect found without balancing that weakens or loses statistical significance after balancing will rarely be credible.¹¹

2.7. Parallel Trends

The parallel-trends assumption—that treated and control units would have experienced parallel changes during the treatment period but for the treatment—is not directly testable, but one during the treatment period can assess plausibility by checking for parallel trends during the pretreatment period. One can graph mean outcomes over time for each group and also provide a *leads-and-lags graph*, which shows the estimated treatment effect over time (see Atanasov and Black 2016 for details). These graphs can be highly revealing both for whether pretreatment trends are parallel and for whether the treatment effect appears when it should relative to the shock. Nonparallel trends are often fatal to a DiD design because one cannot know whether these trends would have continued, stopped, or even reversed but for the treatment (Atanasov and Black 2016). The same is true for shock-IV.

In an IV design, one can provide leads-and-lags graphs for both stages—there should be preshock parallel trends in both the instrumented variable in the first stage and the outcome variable in the second stage. If pretreatment trends in either variable are not parallel, the design is suspect and often cannot be rescued. An IV where the shock is interacted with firm characteristics is similar to a triple-difference design. Parallelism should be checked for both the double differences and the third difference.

3. Reconsideration of Duchin et al. (2010) and Related Board Independence Papers

We turn next from theory to our reexamination of DMO and related papers (this section), Iliev (Section 4), and D&D (Section 5). Unless otherwise specified, all regressions use standard errors clustered on firm. These papers vary in whether they report standard errors or *t*-statistics; we report *t*-statistics throughout.

We study DMO as an example of a set of board independence papers that use one or more of three instruments: *Audit Committee IV*, *Minority-Independent IV*, and *Distance-to-Compliance IV*. We assess these three IVs. We focus in the text principally on DMO's preferred instrument, *Audit Committee IV*, and on threats to validity caused by lack of pretreatment balance. We discuss the other two instruments in more detail in the online appendix, as well as additional concerns with the DMO design.

3.1. Overview of DMO and Audit Committee IV

DMO examine two research questions: (1) does a change in the proportion of independent directors on company boards causally affect three outcomes over 2000–2005 (Tobin's *q*, return on assets (ROA), and stock returns)?, and (2) does the effect of independent directors vary with the firm's informational environment? They

hypothesize that adding independent directors can improve performance in firms with low information acquisition costs yet be counterproductive for firms with high information acquisition costs.

DMO recognize that firms endogenously choose board composition. They rely for causal inference on a legal shock to *audit committee* independence: a 1999 change in NYSE and NASDAQ listing rules that forced listed firms to have 100% independent directors on the audit committee. Previously, a majority-independent committee was permitted. These requirements were later included in the SOX. DMO hypothesize that this legal change was an exogenous shock unrelated to other factors that affect firm choice of board composition, which caused some firms to add independent directors to their boards to staff the audit committee. This rule change affected only firms that lacked a fully independent audit committee in 2000 (the rule became effective after the spring 2000 shareholder meeting season, when most firms elected their 2000 boards). Firms with fully independent audit committees serve as controls.

This is an *encouragement* or *intent-to-treat* design, where the rule change encourages firms with less than 100% independent audit committees to add independent directors to their boards but does not force them to do so. DMO exploit the shock using IV; one could also use DiD. Their core predictor variable is the *change* in percentage of independent directors from 2000 to 2005 ($\delta Indep$). Their instrument for $\delta Indep$ is a dummy variable that equals one for firms without a fully independent audit committee in 2000 (*noncomply dummy*). They use 2SLS and a first-differences specification using differences in the outcomes between 2000 and 2005 and *PctIndep* and covariates measured in 2000. Their covariates are board size, leverage, firm age, $\ln(\text{market capitalization})$, and 48 Fama–French industry fixed effects (FEs).

DMO find that, consistent with prior literature, instrumented $\delta Indep$ does not significantly predict their outcome variables. Their main contribution, other than using a shock-IV design, is to test whether the effect of adding independent directors depends on the firm's information environment measured using an information cost index (*InfoCost*), which combines measures of the number of analysts, analyst forecast dispersion, and analyst forecast error. They find that when *InfoCost* is low (high), adding independent directors adds (subtracts) value for all three of their outcome variables.

3.2. Covariate Balance and Common Support

A strength of the DMO paper is that the authors report on covariate balance in 2000 (DMO, table 2). We present a similar covariate balance analysis for the

DMO sample in the first four columns of Table 1.¹² Table 1 includes several variables that are in the DMO data set but were not used in their analysis. The last three columns assess balance for a trimmed, matched sample (see Section 3.6 for sample details). This sample has good balance on all variables except the treatment dummy.

For the DMO sample, treated and control firms are fairly similar on the DMO outcomes, the covariates that DMO use in their regressions, and the additional potential covariates in their data set. There is moderate imbalance on *InfoCost*. There is, of course, huge imbalance in the percentage of independent directors on the audit committee by construction of their treated and control groups.

More troubling, there is severe imbalance in the preshock percentage of independent directors (*PctIndep*₂₀₀₀). Control firms average 70% independent directors versus 53% for treated firms ($t = 14.81$). This is expected: firms with fewer independent directors are less likely to have fully independent audit committees. However, it is nonetheless cause for

concern. *PctIndep*₂₀₀₀ is an important endogenous firm characteristic that strongly predicts δIndep and may also predict outcomes—indeed, DMO's goal is to assess how *PctIndep* predicts their outcomes. Imbalance on such a covariate, unless corrected, is a recipe for omitted variable bias.¹³

We illustrate graphically imbalance on *PctIndep*₂₀₀₀ and the correlation between *PctIndep*₂₀₀₀ and δIndep (their instrumented variable) in Figure 1. Figure 1(a) shows a scatter plot of *PctIndep*₂₀₀₀ versus δIndep for the DMO sample, with triangles for treated firms and circles for control firms. Only three control firms have *PctIndep*₂₀₀₀ < 25%; only a handful of treated firms have *PctIndep*₂₀₀₀ > 80%. A standard response to lack of common support in the causal inference literature is to trim the sample to a region of reasonably thick common support and study only firms within that region. Any results using the full sample not found for the trimmed sample are suspect because they rely on extrapolation to compare dissimilar treated and control firms. Our judgment is that (25%, 80%) are reasonable trimming bounds.¹⁴

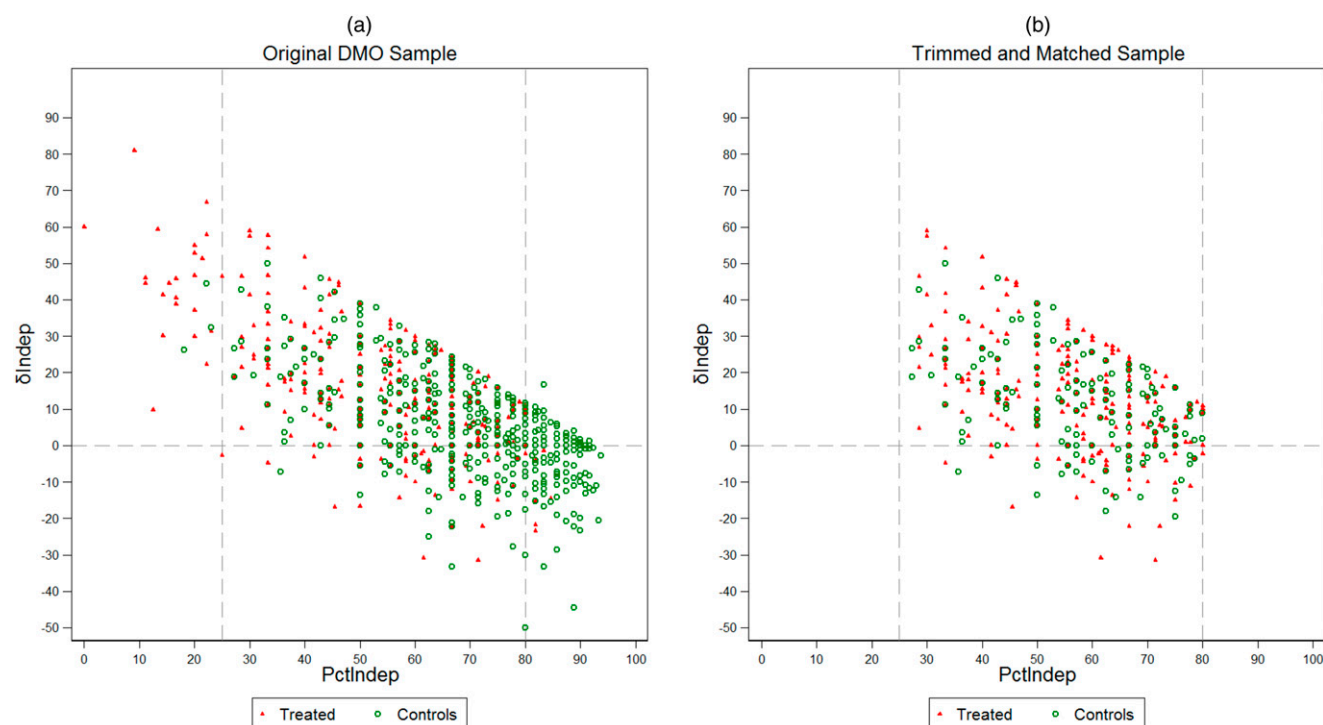
Table 1. Covariate Balance in 2000: Original Sample and Trimmed and Matched Sample

Variable	DMO sample			Trimmed and matched sample		
	Mean (controls)	Mean (treated)	<i>t</i> -test	Mean (controls)	Mean (treated)	<i>t</i> -test
Core variables						
<i>Pct. Independent Directors</i>	69.59	52.73	14.81***	55.26	55.97	0.60
<i>Pct. Independent Directors on Audit Comm.</i>	100.00	63.38	37.54***	100.00	65.59	37.45***
<i>Information Cost Index</i>	0.486	0.459	2.01**	0.467	0.468	0.08
<i>Number of Analysts</i>	16.076	16.683	0.78	15.712	16.974	1.31
<i>Analyst dispersion</i>	0.085	0.067	2.50**	0.001	0.001	0.07
<i>Analyst Forecast Error</i>	0.162	0.152	0.52	0.001	0.002	0.97
Pretreatment outcome variables						
<i>Return on assets (ROA)</i>	0.152	0.154	0.25	0.159	0.150	1.12
<i>Tobin's q</i>	2.215	2.402	1.10	2.394	2.396	0.01
<i>Annual Return</i>	0.013	0.013	0.04	0.009	0.012	0.93
Covariates used by DMO						
<i>log(MV of equity)</i>	7.65	7.81	1.42	7.81	7.85	0.30
<i>Board size</i>	9.63	10.01	1.79	10.02	10.03	0.04
<i>Book leverage</i>	0.408	0.317	1.19	0.377	0.386	0.39
<i>Firm Age</i>	26.54	25.68	0.80	25.80	26.62	0.61
Other available variables (in DMO data set)						
<i>ln(Tobin's q)</i>	0.561	0.605	1.01	0.618	0.603	0.28
<i>Total Assets</i>	10,847	15,689	1.17	14,371	16,520	0.36
<i>ln(Assets)</i>	7.73	7.81	0.71	7.78	7.87	0.62
<i>Annualized std. dev. of returns</i>	0.152	0.145	1.38	0.150	0.144	1.17
<i>Intangible assets</i>	0.719	0.700	1.17	0.717	0.696	1.03
<i>Market/book ratio</i>	3.597	3.836	0.64	4.427	3.909	0.88
<i>Number of business segments</i>	2.908	2.808	0.73	2.559	2.819	1.57
Number of unique firms	582	323		267	267	
After trim to common support, before matching				343	267	

Notes. The table shows covariate balance for treated and control firms in 2000. Treated (control) firms lack (have) 100% independent directors on the audit committee in 2000. Columns under DMO sample use DMO sample (905 firms included in Tobin's q regressions in Table 2). Columns under "Trimmed and matched sample" are trimmed to common support (trim to *PctIndep*₂₀₀₀ ∈ (25%, 80%]; estimate propensity score p ; trim to common support on p ; reestimate p) followed by one-to-one matching on propensity score with replacement. The one-to-one matched sample is 267 treated firms and 267 matched control firms (156 unique firms). The table shows two-sample t -tests for differences in means. Significant results (at 5% or better) are in bold. Market value of equity and total assets are in millions of dollars.

Significance at the 5% level; *significance at the 1% level.

Figure 1. (Color online) Scatter Plots of $PctIndep_{2000}$ vs. $\delta Indep$



Notes. Scatter plots of percentage of independent directors in 2000 ($PctIndep$) and change in this percentage from 2000 to 2005 ($\delta Indep$). Treated (control) firms are firms that do not have (have) 100% independent audit committee in 2000. Treated (control) firms are shown with triangles (circles). We add vertical lines at $PctIndep_{2000} = 25\%$ and 80% to highlight the limited overlap between treated and control firms outside these bounds. (a) DMO sample = 905 firms used in Tobin's q regressions in Table 2 (323 treated; 582 controls). (b) Trimmed and matched sample (267 treated firms and 267 matched control firms using one-to-one matching on propensity score with replacement; 156 distinct control firms). Correlation between $PctIndep_{2000}$ and $\delta Indep$ is $r = -0.67$ for DMO sample and -0.52 for trimmed and matched sample.

It is also apparent that $PctIndep_{2000}$ strongly predicts $\delta Indep$. Firms with fewer (more) independent directors in 2000 add more (fewer) independent directors by 2005. The overall correlation between $PctIndep_{2000}$ and $\delta Indep$ is -0.69 ($t = 27.4$), and it is similar for treated and control firms.¹⁵ This is evidence of regression to the mean. Firms with low $PctIndep_{2000}$ (mostly treated) increased board independence relative to firms with high $PctIndep_{2000}$ (mostly control firms). Unless one ensures balance on $PctIndep_{2000}$, this will produce strong OVB in the first stage, which will ascribe the greater increase in board independence among treated firms to the IV, even though most treated firms with low $PctIndep_{2000}$ would have increased their board independence for other reasons, as shown by the negative slope of the data cloud for control firms.

This regression-to-mean concern is quite general. It is likely to apply any time (1) a shock is based on (or for *Audit Committee IV*, related to) whether a firm is above or below a threshold and (2) firms tend to move toward the threshold without the shock. These factors will often be present when the threshold is a percentage, where firms with a low (high) initial percentage are likely to increase (decrease) that percentage.¹⁶

Third, for the DMO sample, consider vertical slices in Figure 1, for which treated and control firms have similar $PctIndep$. If *Audit Committee IV* strongly induced firms to add independent directors to their boards, the triangles should be systematically above the circles. Triangles and circles are instead well mixed. This is evidence that *Audit Committee IV* has little power for firms with similar $PctIndep_{2000}$.

If one does not ensure pretreatment balance on $PctIndep_{2000}$, there will be OVB in the second stage as well if $PctIndep_{2000}$ predicts the DMO outcomes either directly or through a channel other than *Audit Committee IV*. Unless these channels can be ruled out on theoretical grounds—and in the board independence studies, they cannot—credible causal inference requires balance on $PctIndep_{2000}$.

Figure 1(b) uses the trimmed and matched sample. Within the common support region, treated and control firms are well mixed, suggesting that *Audit Committee IV* has limited power to affect board independence.

We investigate the limited overlap between treated and control firms on $PctIndep_{2000}$ further in Figure 2(a). We divide the DMO sample into 20 bins based on $PctIndep_{2000}$, covering $[0\%–5\%]$, $(5\%–10\%]$, $(10\%–15\%]$,

etc., and we show the number of treated and control firms in each bin. There are no control firms with $PctIndep_{2000} < 15\%$, and only a few with $PctIndep_{2000} < 25\%$. In our judgment, sensible trimming bounds for DMO are $PctIndep_{2000} \in (25\%, 80\%]$. Trimming at 25% excludes 24 treated firms but only three control firms; trimming at 80% excludes only six treated firms but 153 control firms.¹⁷ Figure 2(b) is similar but for the trimmed and matched samples; this sample has common support (by construction). Unlike the DMO sample, it is balanced on $PctIndep_{2000}$, which is one of the variables we use in matching.

In Figure 3(a), we examine shock strength for firms within these trimming bounds using the same bins for $PctIndep_{2000}$ as in Figure 2. We show mean $PctIndep_{2000}$ separately for treated and controls within each bin. The dashed lines in Figure 3 show the overall sample means for $\delta Indep$ for treated and controls. The overall difference in means for the trimmed DMO sample is 7%, well below the 11% first stage in Table 2; this reflects trimming to common support. Most within-bin differences are small. These small differences confirm the impression from Figure 1 that after one controls for $PctIndep_{2000}$, Audit Committee IV loses much of its power to predict $\delta Indep$.

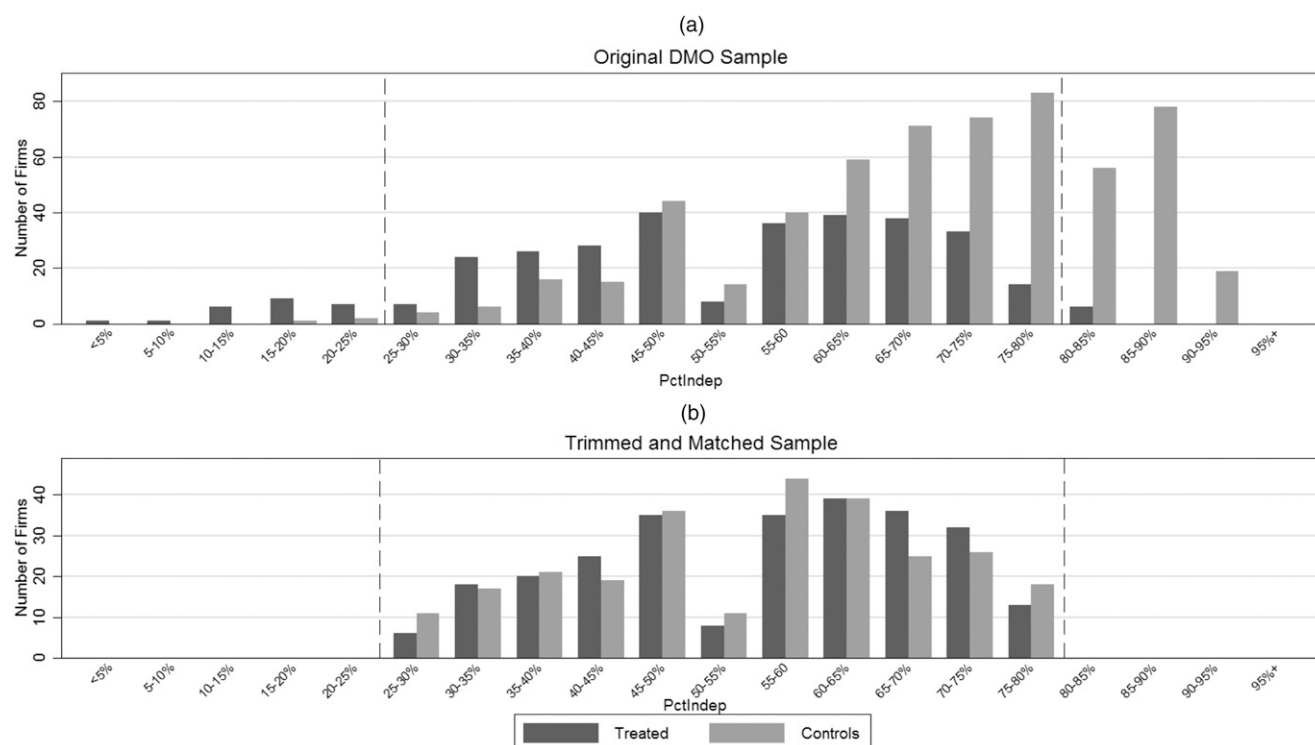
Figure 3(b) is similar but for the trimmed and matched samples. The average difference $\delta Indep$ between treated and control firms is now only 1.8%. This small difference implies a correspondingly weaker first stage in a 2SLS analysis; the remaining first stage will turn out to be too weak to be usable.

3.3. DMO Results Correcting for Technical Errors

We begin our analysis by correcting two technical errors that DMO make. We discuss these errors only in a note because Duchin et al. (2015) agree that these were errors. See the online appendix for additional details.¹⁸

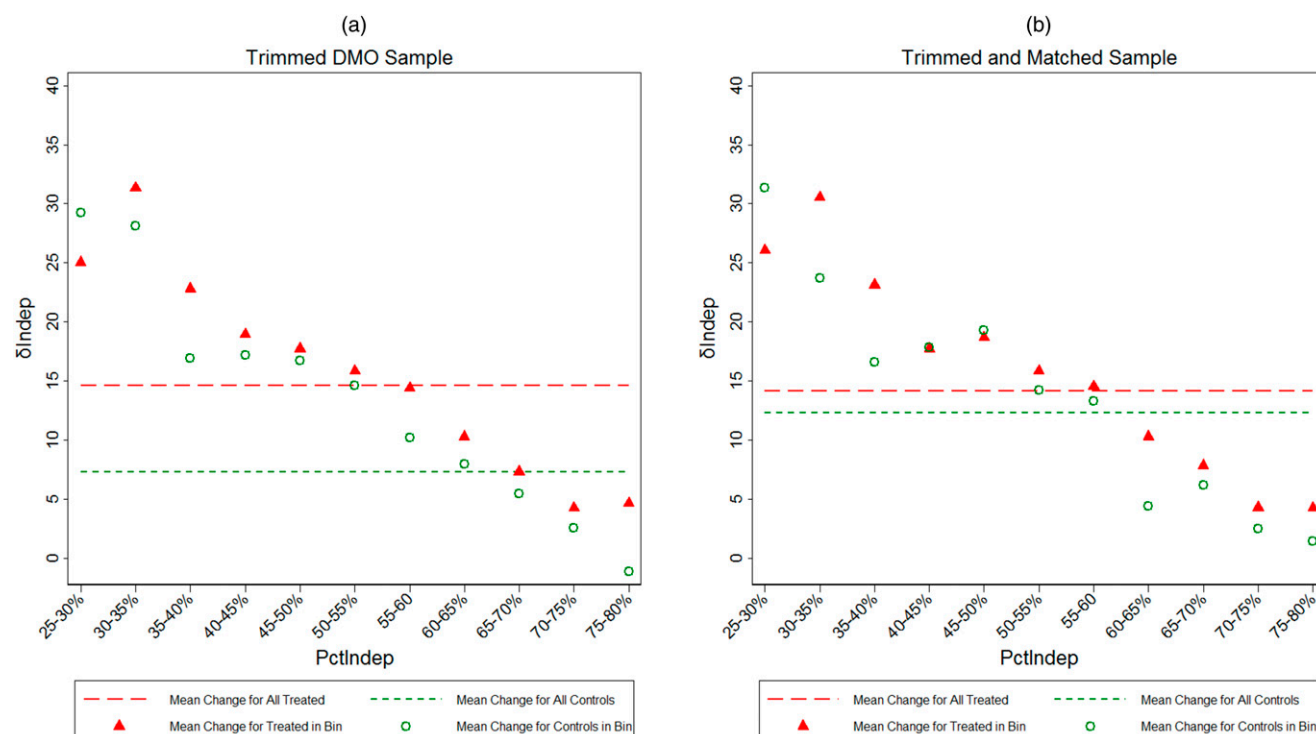
Table 2 reports results using the DMO model and sample after correcting these errors. We report first-stage results in columns (1) and (2) and second-stage results in columns (5)–(7) in Table 2. (We discuss the remaining columns later.) The corrected dual IVs are substantially weaker than DMO's single IV but remain acceptably strong. In the second stage, the coefficients on their core interaction term between instrumented $PctIndep$ and instrumented $InfoCost$ are substantially smaller than the results that DMO report. The coefficient for δROA is only marginally significant; the average t -statistic across all three outcomes falls from 5.33 to 2.24. For the other two IVs,

Figure 2. Number of Treated and Control Firms Within $PctIndep_{2000}$ Bins



Notes. Histogram plots of the number of treated and control firms by percentage of independent directors in 2000 ($PctIndep$). We add vertical lines at $PctIndep = 25\%$ and 80% to highlight the limited overlap between treated and control firms outside these bounds. Firms with exactly 25% independent directors are included in the 20%–25% bin, and this is similar for other bins. Samples are the same as in Figure 1. Vertical dashed lines indicate regions with reasonable common support ($PctIndep_{2000} \in (25\%, 80\%]$).

Figure 3. (Color online) $\delta Indep$ for Treated and Control Firms within Bins for $PctIndep_{2000}$



Notes. The figure shows mean $\delta Indep$ separately for treated and control firms within bins for $PctIndep_{2000}$. Firms with exactly 30% independent directors are included in the 25%–30% bin, and this is similar for other bins. (a) Trimmed DMO sample: 719 firms (297 treated, 422 control) with $PctIndep_{2000} \in (25\%, 80\%]$. For this sample, the mean treated minus control difference in $\delta Indep$ is 7.33%. For firms with $PctIndep_{2000} \leq 50\%$, this difference is 3.85%. (b) Trimmed and matched sample is the same as in Figure 1. For this sample, the mean treated minus control difference $\delta Indep$ is 1.83%, and it is 1.30% for firms with $PctIndep_{2000} \leq 50\%$.

second-stage statistical and economic significance disappears with these corrections (see Table DMO-App-1B in the online appendix).

3.4. Theoretical Need to Balance on $PctIndep$

We show below the problems that arise in the DMO design if one does not ensure balance on $PctIndep$, but we begin here with theory. Why should one balance on $PctIndep$? First, $PctIndep$ strongly predicts Audit Committee IV, could also predict DMO's outcomes, and does so in their theory. From classic regression theory, omitting a variable that predicts both the instrument and the outcome will generate OVB. Causal inference theory also supports balancing on extensive, potentially relevant pretreatment covariates to bring assignment to treatment closer to being as if randomly assigned.

Another reason for balancing on $PctIndep$ builds on the need for shock-IV designs to satisfy the parallel-trends assumption. In Figure 4(A1), we plot simple annual means for $PctIndep$ by year for the treated and control firms. Pretreatment trends are strongly non-parallel. $PctIndep$ rises for control firms from 64% in 1996 to almost 70% in 2000, yet it falls for control firms from 55% in 1996 to 53% in 2000.¹⁹ This problem

disappears for the trimmed and matched sample (Figure 4(A2)).

3.5. Problems with the DMO Design Without Balance on $PctIndep$

In the balance of this section, we first illustrate what can go wrong when pretreatment balance is absent by *not* enforcing balance on $PctIndep$ (the approach that DMO advocate) and running a battery of diagnostic and falsification tests drawn from IV, DiD, and RD designs. We show that the Board Independence IVs do not pass these diagnostics. We seek to illustrate both what can go wrong if covariate balance and common support are absent and the importance of falsification tests.

We then impose balance. We trim the sample to common support and use propensity score matching to ensure pretreatment covariate balance. We rerun the DMO analysis on the trimmed and matched sample and show that DMO's first-stage (and second-stage) results for *Audit Committee IV* disappear. The other two instruments are not amenable to matching, but we show in the online appendix, Figures DMO-App-1 and DMO-App-5, that these IVs have no power to predict a change in board independence,

Table 2. 2SLS with Audit Committee IV Correcting DMO Technical Errors for Original and Matched Sample

	First stage				Second stage (2SLS)					
	DMO sample		Trimmed and matched sample		DMO sample			Trimmed and matched sample		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dependent variable	$\delta Indep$	$\delta Indep \times InfoCost$	$\delta Indep$	$\delta Indep \times InfoCost$	δROA	$\delta \ln(Q)$	Mean return	δROA	$\delta \ln(Q)$	Mean return
<i>Noncomply dummy</i>	10.490*** (3.23)	−0.299 (0.21)	−4.378 (−0.94)	−2.295 (−1.18)						
<i>Noncomply dummy × InfoCost</i>	2.869 (0.46)	12.482*** (3.49)	12.855 (1.41)	6.851 (1.35)						
<i>Instrumented $\delta Indep$</i>					0.234* (1.72)	1.025 (1.63)	0.063*** (3.20)	−5.354 (−0.39)	−60.670 (−0.08)	−0.734 (−0.42)
<i>Instrumented ($\delta Indep \times InfoCost$)</i>					−0.507* (1.72)	−2.758** (2.15)	−0.116*** (2.85)	9.723 (0.36)	110.214 (0.08)	1.382 (0.37)
<i>InfoCost</i>	−4.025 (0.87)	3.480 (1.42)	−5.072 (−0.81)	10.376*** (2.91)	−0.914 (0.29)	−28.171* (1.89)	0.049 (0.11)	−134.247 (−0.37)	−1487.707 (−0.08)	−18.678 (−0.39)
<i>Board Size</i>	−0.127 (0.64)	−0.122 (1.18)	−0.530** (−2.05)	−0.292* (−1.85)	−0.039 (0.27)	1.060 (1.49)	−0.014 (0.72)	−0.058 (−0.06)	1.597 (0.16)	0.023 (0.08)
<i>Book Leverage</i>	0.237 (0.54)	0.044 (0.17)	9.833** (2.09)	5.134* (1.83)	0.940*** (2.67)	4.835*** (5.83)	0.029 (0.49)	11.530 (0.87)	53.397 (0.18)	1.265 (0.43)
<i>Age</i>	−0.073* (1.89)	−0.015 (0.81)	−0.109* (−1.78)	−0.037 (−1.34)	0.021 (0.97)	0.617*** (4.95)	0.006* (1.69)	−0.210 (−0.40)	−2.006 (−0.07)	−0.021 (−0.39)
<i>Market Cap</i>	0.191 (0.78)	0.076 (0.55)	0.561 (1.10)	0.087 (0.32)	−0.456*** (3.07)	−15.263*** (7.85)	−0.390*** (8.57)	1.391 (0.31)	8.099 (0.03)	−0.167 (−0.35)
FE industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage <i>F</i> -statistic	49.08	45.38	1.72	0.94	—	—	—	—	—	—
<i>p</i> -value	(0.0000)	(0.0000)	(0.1936)	(0.3987)	—	—	—	—	—	—
No. of firms	905	905	534	534	897	905	805	525	534	476
No. of treated firms	323	323	267	267	321	323	288	262	267	231
Adjusted <i>R</i> ²	0.16	0.17	0.15	0.21	0.04	0.33	0.19	0.08	0.35	—

Notes. DMO replication results with technical corrections (discussed in the text) for the DMO sample and the trimmed and matched sample. DMO Sample is 905 firms included in the Tobin's *q* regressions. Matched sample is trimmed to common support (trim to $PctIndep_{2000} \in (25\%, 80\%]$; estimate propensity score *p*; trim to common support on *p*; reestimate *p*) followed by one-to-one matching on propensity score with replacement. Matched sample is 267 treated firms and 267 matched control firms (156 unique firms). Columns (1) and (2) and (3) and (4) report first-stage results using *noncomply dummy* and *noncomply dummy × InfoCost* as instruments for $\delta Indep$ and $\delta Indep \times InfoCost$. Columns (5)–(10) report the second-stage results. Reported first-stage regressions are for sample with $\delta \ln(Q)$ as outcome variable. The *t*-statistics, with industry clusters, are in parentheses. Significant results (at 5% or better) are in bold.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

once we take into account the regression-to-the-mean relation between *PctIndep* and $\delta Indep$. These IVs also lose first-stage strength if we run 2SLS controlling for *PctIndep*. See Table DMO-App-4C.

3.5.1. Overview of Falsification and Placebo Checks Without Balance on *PctIndep*. DMO did not control for a key covariate, *PctIndep*, which strongly predicts both their instrument (a nonfully independent audit committee) and their outcome $\delta Indep$. We show next that this choice drives their results. Were they right to omit *PctIndep*? How should one assess validity for the Minority-Independent and Distance-to-Compliance IVs, for which balancing on *PctIndep* is not possible, by construction of these IVs?

The Board Independence IVs provide a useful clinic on how IV analysis can go wrong if one lacks pretreatment balance and on the value of placebo and other falsification tests in uncovering the problems. All three IVs exhibit severe nonparallel pretreatment trends for *PctIndep* (and thus $\delta Indep$) between treated and control firms. These nonparallel trends drive a number of other results. Using the DMO specification, which does not control for *PctIndep*, we find for all three IVs that:

1. A *placebo shock-IV* (whether a firm has a nonfully independent audit committee in 1998, two years before the actual shock) has a strong first stage during the pretreatment period of 1998–2000. There is nothing special about 1998—a placebo shock-IV defined in

any pretreatment year strongly predicts $\delta Indep$ during the pretreatment period.²⁰

2. These IVs strongly predict $\delta Indep$ and thus, have a strong first stage looking *backward in time*—predicting from 2000 back to 1996!

These results are impossible for a valid IV. For DMO's research design, there are further issues (see the online appendix).

3. DMO argue in their reply (Duchin et al. 2015) that *Audit Committee IV* should work principally for firms *without* majority independent boards in 2000, because these firms are more likely to need to add independent directors to their boards to staff the audit committee. In fact, *Audit Committee IV* is weak where they say it should be strong. For firms without majority independent boards, this IV lacks first-stage power, even before we impose balance on $PctIndep_{2000}$.

iv. The DMO design, in which they interact the Audit Committee Shock with *InfoCost*, is similar to an intent-to-treat triple-difference (DiDiD) design, with a continuous third difference (Atanasov and Black (2016) call this a DiDiD-continuous design). If we partition their sample based on above- versus below-median *InfoCost*, we find severe nonparallel pretreatment trends for their outcome variables between high and low *InfoCost* firms. This is separately fatal for their design.

v. Their second stage seems to be driven by the 2001 bursting of the internet bubble rather than their posited channel. High *InfoCost* firms had much higher preshock values for ROA and Tobin's q in 2000 near the peak of the bubble; these levels then plummeted in 2001 and 2002.

3.5.2. Regression to the Mean in the DMO Design.

Figure 4(B1) shows the difference in mean $PctIndep$ between the two groups over time both for the actual shock year (2000) and for placebo shocks in 1998 and 1999. The placebo-treated firms are those without fully independent audit committees in the placebo shock years. The V shape for all three years is a classic regression-to-the-mean shape. Both the actual and placebo *Audit Committee IVs* select firms that lagged their peers in $PctIndep$ prior to the shock. Figure 1 provides evidence that these firms would have moved toward those peers in the postshock period even without the shock.

The shapes of all three graphs are similar. This is powerful evidence that regression to the mean drives the apparent power of Audit Committee IV.²¹ Indeed, the V shapes imply that the IV predicts the instrumented variable, $\delta Indep$, looking backward in time. This is fatal for IV validity unless one addresses imbalance on $PctIndep$. We confirm with covariates in Figure DMO-App-2 and Table DMO-App-4 in the

online appendix this impression from the univariate graphs. All three IVs have strong first stages predicting backward in time, with coefficients and t statistics similar to those looking forward in time for similar look-back and look-forward periods. We show the similar predictive power of *Audit Committee IV* looking forward and backwards in time in Figure DMO-App-2A. Figures for the other two instruments are similar and not shown. In Figure DMO-App-2B, we show that the apparent strength of *Audit Committee IV* disappears once we balance and trim sample.

3.6. Addressing Imbalance on $PctIndep_{2000}$ for Audit Committee IV

3.6.1. Trimmed and Matched Sample. Ignoring imbalance on $PctIndep_{2000}$ leads to major problems. We turn next to what one should do—improve pretreatment balance. There are many plausible balancing methods. We pursue one approach in the text—trimming to common support, matching on the propensity score p (propensity to have a non-fully independent audit committee in 2000), using 1:1 matching with replacement, and regression on the matched sample. In our view, it is often good practice to use several balancing approaches and verify whether they produce similar results. In the online appendix, we present results if we trim to common support and then, use several other approaches: propensity score matching without replacement, inverse propensity weighting, subclassification on the propensity score followed by regression within subclasses, and 2SLS regressions with $PctIndep_{2000}$ as a covariate (See Tables DMO-App_7, DMO-App-9, DMO-App-10, and DMO-App-11). All of these approaches produce similar results. Trimming and matching are feasible for *Audit Committee IV* but not for the other two IVs, for which common support does not exist, by construction of these IVs. We discuss in section 3.8 of the online appendix how one might address imbalance on $PctIndep$ for these IVs.

Our trimming and matching proceeds as follows. We (1) trim to $PctIndep_{2000} \in (0.25, 0.80]$ (sample = 293 treated, 426 controls), (2) estimate the propensity to be treated (have a non-fully independent audit committee in 2000) based on all available pretreatment covariates, including $PctIndep_{2000}$ and *InfoCost*, using logit regression, (3) trim to literal common support ($p \in [p_{\min, \text{control}}, p_{\max, \text{treated}}] = [0.09, 0.88]$) (269 treated and 387 control firms), and (4) reestimate propensities and trim again to literal common support. The resulting sample is 267 treated and 343 control firms, with propensities $p \in [0.15, 0.90]$. We then use one-to-one matching with replacement to generate a final sample of 267 treated and 267 control firms (156 unique firms).²²

As shown in the last three columns of Table 1, the resulting sample is well balanced on all covariates and outcomes, including *PctIndep*₂₀₀₀ and *InfoCost*. It remains imbalanced only on the treatment dummy. We show graphical results for the trimmed, matched sample in Figures 1(b), 2(b), and 3(b). In Figure 1(b), the imbalance on *PctIndep*₂₀₀₀ seen in Figure 1(a) has disappeared. In Figure 2(b), the imbalance in number of treated and control firms seen for the DMO sample even within the common support region (most treated firms have low *PctIndep*₂₀₀₀; most control firms have high *PctIndep*₂₀₀₀) has disappeared.

The matching process has done its job; assignment to treatment is no longer significantly related to *PctIndep*₂₀₀₀. Figure 3(b) also reflects improved balance on *PctIndep*₂₀₀₀, with only a 1.8% average difference between treated and control firms (1.3% for firms without majority-independent boards—the firms for which DMO assert that Audit Committee IV should have power). Although Audit Committee IV is weaker, it does not display the pathologies that we saw in Figure 4. The non-parallel trends in Figure 4(A1) and the V shapes in Figure 4(B1) disappear.²³

3.6.2. Results: No First Stage. We show 2SLS results for the trimmed and matched sample in Table 2. The combined instruments become statistically insignificant (columns (3) and (4)).²⁴ This result was foreshadowed by the small treated minus control differences in mean $\delta Indep$ in Figure 3B. The second stage disappears as well (columns (8)–(10) in Table 2); indeed, the signs on the core interaction terms that DMO study flip and become positive for all three outcomes.

Loss of sample size from trimming and matching will, of course, directly reduce power. However, the loss of sample size cannot explain the differences in results between this sample and the original DMO sample. We use simulation to assess the effect of sample size on power by drawing 1,000 samples of 534 firms at random from the original DMO sample and rerunning the DMO model (not balancing on *PctIndep*); we report results in Table DMO-App-5 in the online appendix. The DMO instruments retain reasonable power (mean *F*-statistics of 22 and 23).

3.7. Assessing Preshock Parallel Trends

DMO use data only for 2000 and 2005, compute outcomes as differences between 2000 and 2005, measure covariates in 2000, and then run cross-sectional regressions with industry clusters. An alternative is a true panel data design with firm FEs, annual data, and standard errors clustered on firm. With full panel data, one can use leads-and-lags graphs

to plot annual coefficient estimates for both stages; this can be very informative both for whether one has parallel pretreatment trends and whether the results in each stage appear when they should relative to the shock.²⁵ If results take time to emerge, one can capture the full effect of the shock using distributed lag regressions (Atanasov and Black 2016, section 4.4).

We provide leads-and-lags results for ROA and Tobin's *q* in Figure DMO-App-3 in the online appendix. These show nonparallel trends for samples partitioned on either *Audit Committee IV* or *InfoCost*. This provides evidence that their IV is not as if randomly assigned. This strong nonparallelism would be fatal for the DMO design, even if they had a strong instrument.

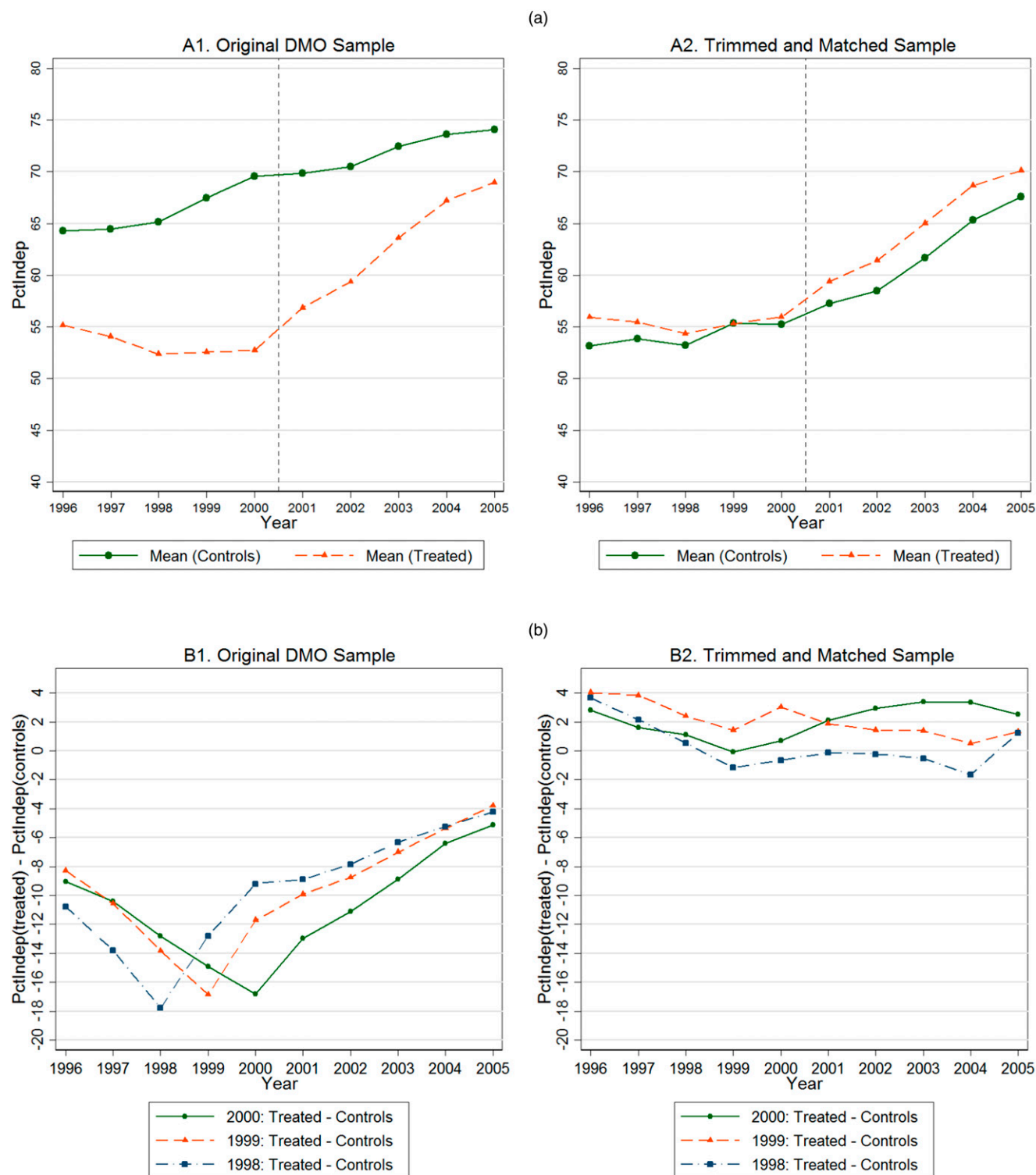
3.8. Summary for Board Independence IVs and Further Comments

3.8.1. Covariate Imbalance on *PctIndep* and Regression to the Mean. The Board Independence IVs have extreme imbalance on a core preshock covariate (*PctIndep*₂₀₀₀). That imbalance plus regression to the mean drives the apparent strength of all three instruments. These instruments fail placebo checks: placebo instruments predict strongly in the pretreatment years, and both the actual and placebo instruments predict backward in time about as strongly as forward in time. After we balance on *PctIndep*₂₀₀₀, which is feasible for *Audit Committee IV*, the first stage loses statistical strength. The nudge toward higher board independence that this IV provides is too weak to be usable.

3.8.2. Covariate Imbalance and Parallel Trends for Third Difference. For an IV that relies on an interaction, one should check covariate balance for the third variable (for DMO, *InfoCost*) and confirm parallel pretreatment trends for the double differences (for *PctIndep* and *InfoCost*) as well as the triple difference. The DMO sample has strong covariate imbalance for firms with high versus low *InfoCost*. In the online appendix, Figures DMO-App-3E and DMO-App-3F, we show that the DMO sample exhibits sharply nonparallel trends for subsamples partitioned on *InfoCost* $\times$ *Audit Committee IV*. That imbalance seems to drive their second-stage results.

3.8.3. Only-Through Condition. The only-through condition requires that an increase in audit committee independence affects the DMO outcome variables only through board independence. One must believe that the shock to audit committee composition has no direct effect on ROA, Tobin's *q*, or share returns. However, why should this be true? Fully independent audit committees may (1) affect firms' accounting choices and hence reported profits, which would

Figure 4. (Color online) Parallel Trends Graph for Percentage Independent Directors



Notes. Panels (A1) and (B1) use the same sample as in Table 2 (323 treated; 582 controls). Panels (A2) and (B2) use the trimmed and matched sample (267 treated and matched controls, controls include 156 distinct firms). Panel (a) shows annual means for *PctIndep* over 1996–2005 separately for treated firms (without fully independent audit committees in 2000) and control firms. Dashed vertical lines separate the pre-treatment and posttreatment periods. Panel (b) shows annual differences in means for the actual shock in 2000 and placebo shocks in 1998 and 1999 defined using audit committee independence in these years instead of 2000. The similar shapes of the three lines provide evidence that regression to the mean rather than the audit committee shock drives the V shapes.

affect ROA, or (2) reduce fraud risk and thus lead investors to pay more for the same reported earnings, which would affect Tobin's q and share returns. D&D and Iliev address the only-through condition with care; DMO simply note that other channels are possible.

One reason why we like to see both intent-to-treat DiD and shock-IV results in the same paper is that intent-to-treat DiD avoids this assumption. A DiD analysis would let one assess whether the Audit Committee Shock caused a change in their outcome variables while remaining agnostic on what channel causes the effect—a direct effect of the audit committee change, an indirect effect through board independence, or a third channel (perhaps appointing directors with financial expertise to staff the audit committee).

3.9 The DMO Theory Could Still Be Right

The DMO theory—that independent directors are less effective if they face high costs to acquire firm-specific information—could still be right. The Board Independence IVs are simply too weak to let them test this theory.

4. Reanalysis of Iliev (2010)

4.1. Overview

SOX section 404, as implemented by the SEC and later amended by the Dodd–Frank Act, requires firms with *public float* (the market value of shares not held by insiders) $> \$75\text{M}$ to have their auditors certify that the firm has adequate internal controls beginning in 2004. Complying with SOX section 404 surely increases firms' auditing costs, but one would like to know by how much. For larger public firms, there is no control group, and therefore, the best available research design is interrupted time series, in which one would estimate the time trend in audit fees before and after 2004 and look for an unusual jump in 2004 relative to that trend. However, audit fees might have changed in 2004 for reasons other than SOX section 404.

In a careful and clever paper, Iliev (2010) seeks to do better for firms near the \$75M free float threshold for SOX section 404 to apply. His central research design is RD, in which he compares firms just above this threshold with firms just below it. He estimates a mean increase in $\ln(\text{audit fees})$ of 0.744 ($t = 7.39$). The corresponding percentage increase is $e^{0.744} - 1 = 110\%$, implying that audit fees more than doubled.

Iliev finds evidence that some firms manipulate their free float to avoid complying with SOX section 404. He argues that these firms might have especially high SOX compliance costs, which would bias his estimate of average costs downward. He addresses this concern with an IV strategy. The SEC adopted the rule containing the \$75M threshold for SOX compliance in 2003. The rule required firms to comply in 2004

if their float in 2002, 2003, or 2004 exceeded \$75M. Firms could not go back in time to manipulate their 2002 float, and in 2002, they had no reason to expect \$75M in float to become a magic number in the future. Iliev therefore uses 2002 float $> \$75\text{M}$ as an instrument for SOX compliance. His 2SLS estimated increase in $\ln(\text{fees})$ is 0.983 ($t = 3.65$). The corresponding percentage increase is $e^{0.983} - 1 = 167\%$, well above his OLS estimate. See Table Iliev-App-2, regression (4) in the online appendix.

Iliev (2010) is a delightful and careful paper. In Atanasov and Black (2016), we review 863 empirical corporate governance papers published in 22 major journals over 2001–2011 and find 74 shock-based papers. Iliev (2010) is one of only two RD papers in our sample. He checks for covariate balance, includes a flexible control for the running variable (float or, more generally, firm size), varies the bandwidth around the discontinuity,²⁶ checks for evidence that firms manipulate their float to avoid SOX compliance, and runs an array of placebo and robustness checks. He finds evidence of threshold manipulation and addresses it with a clever and plausible instrument. His RD design is careful and credible.

Our focus here, however, is on Iliev's use of shock-IV. His IV is creative, is clearly exogenous, and at first glance seems impregnable. Yet, appearances are deceiving. We show below that his RD design, for subtle reasons, produces covariate balance in 2002 when SOX was adopted but no longer produces balance on firm growth by 2004 when SOX section 404 comes into effect. This alone does not matter much. Controlling for growth modestly reduces the coefficient on the SOX compliance dummy from 0.744 to 0.706; our preferred RD/DiD design provides an estimated coefficient of 0.586 (an 80% increase in audit fees).

Even more subtly, Iliev's IV design leads to severe imbalance on growth between complier and control firms. We show below that his IV turns out to predict audit fees both through the instrumented variable (SOX section 404 compliance) and through omitted growth variables—a violation of the only-through condition. The *IV compliers* (who comply with SOX section 404 only because they are forced to do so by his instrument) are also a small, likely unrepresentative subsample of all firms that must comply with SOX (call these *SOX compliers*). We also show that the concern that led Iliev to use IV—that firms that manipulated their float to avoid SOX section 404 might have relatively high SOX compliance costs—is not a significant issue for his study.

4.2. Data Manipulations

We start with Iliev's data set, which includes 1,492 unique firms over 2002–2004 (the period that we focus on here). Of these, 815 firms have full data on float,

audit fees, and covariates, and 281 are within his [\$50M, \$100M] bandwidth for 2004 float. In the online appendix, Tables Iliev-App-1 and Iliev-App-2, we replicate his RD and IV results.

In our reanalysis, we vary bandwidth systematically in both 2002 and 2004. To do so, we define a bandwidth parameter b and a corresponding bandwidth of $[\frac{\$75M}{b}, \$75M \times b]$. In our main analysis, we use $b = 1.5$, which implies a bandwidth of [\$50M, \$112.5M], similar to Iliev's bandwidth.

4.3. Covariate Balance

We assess covariate balance in Tables 3–5 both in 2002 (before the \$75M threshold was established) and in 2004 (when Iliev builds his sample). Iliev assesses balance in 2002 but not in 2004. In Table 3, we compare future-treated firms (2002 float $\in$ [\$75M, \$100M]) with potential future control firms (2002 float $\in$ [\$50M, \$75M]). The two groups are similar on all of Iliev's covariates except size and related variables. In 2004 (Table 4), treated firms are similar to control firms on Iliev's covariates. So far, so good.

However, are the SOX complier and control firms really similar? In both tables, we also include variables for growth from 2002 to 2004 in float, market capitalization, sales, and assets. In 2002, below- and above-threshold firms are similar except for size and related variables, which they should be. However, float and market cap grew much more slowly for SOX compliers than for controls (Table 4). These

differences arise because the SEC rule is asymmetric on how changes in float affect compliance. Firms can grow into compliance but cannot shrink out of compliance.

A core RD assumption is thus violated. Iliev missed this. So did we until we carefully thought about which firms were the IV compliers, always takers, and never takers. Does omitting growth covariates matter? The answer, for Iliev's RD results, is "not much." In Table App-Iliev-1, regression (2A) in the online appendix, we add growth variables to his regression. The treatment effect estimate drops from 0.744 to 0.706. However, imbalance on growth will become central to our reassessment of his IV strategy. The IV compliers (the firms that comply only because they have 2002 float $>$ \$75M) are highly unbalanced versus the controls; much more so than all SOX compliers.

4.4. Impact of Growth Imbalance on Coefficient Estimates

We next investigate the effect of imbalance in growth on estimates of SOX section 404 compliance costs, relying primarily on graphical analysis. Note first that the *IV compliers*—the firms that comply with SOX section 404 only because they had float $>$ \$75M in 2002—are a particular subgroup of all *SOX compliers*. The IV compliers must have had float $>$ \$75M in 2002 yet float $<$ \$75M in each of 2003 and 2004. Could this negative growth trajectory explain their 2004 audit fees, at least in part?

Table 3. Summary Statistics for Covariate Balance in 2002 for 180 Firms in the Iliev Data Set with 2002 Free Float $\in$ [\$50M, \$75M] vs. 110 Firms with 2002 Free Float $\in$ [\$75M, \$100M]

Variable	Means			
	Float $<$ \$75M	Float $>$ \$75M	Norm. difference	<i>t</i> -statistic
Size variables				
ln(float)	4.474	4.769	0.194	2.14**
ln(market cap)	4.819	5.027	0.277	3.34***
ln(sales)	4.173	4.527	0.107	1.28
ln(assets)	4.646	4.982	0.238	2.86***
Iliev's other covariates				
Leverage	0.135	0.159	0.086	1.02
Receivables/assets	0.257	0.298	0.132	1.57
Big auditor	0.783	0.791	0.013	0.15
No. of business segments	1.739	1.845	0.057	0.68
No. of geographic segments	1.600	2.145	0.229	2.77***
Growth variables				
Change in ln(float)	0.353	0.315	−0.025	0.28
Change in ln(market cap)	0.252	0.124	−0.117	1.30
Change in ln(assets)	0.121	0.158	0.057	0.68
Change in ln(sales)	0.134	0.159	0.026	0.31

Notes. Growth variables are measured from 2002 to 2004. The table shows *t*-statistics for differences in means between treated and control groups and *normalized differences* (defined in Table 8). Significant differences (at 5% level or better) are in bold.

Significance at the 5% level; *significance at the 1% level.

Table 4. Summary Statistics for Covariate Balance in 2004 for 164 SOX Compliers and 117 Controls with 2004 Free Float $\in$ [\$50M, \$100M]

	Means			
Variable	Controls	SOX compliers	Norm. difference	<i>t</i> -statistic
Size variables				
<i>ln(float)</i>	4.128	4.386	0.762	13.43***
<i>ln(market cap)</i>	4.385	4.783	0.533	7.42***
<i>ln(sales)</i>	4.221	4.138	−0.027	0.31
<i>ln(assets)</i>	4.467	4.701	0.154	1.855*
Iliev's other covariates				
<i>Leverage</i>	0.177	0.152	−0.085	1.00
<i>Receivables/assets</i>	0.304	0.261	−0.141	1.67*
<i>Big auditor</i>	0.744	0.829	0.146	1.75*
<i>No. of business segments</i>	2.154	1.774	−0.182	2.19**
<i>No. of geographic segments</i>	1.974	1.976	0.000	0.01
Growth variables				
<i>Change in ln(float)</i>	0.960	0.303	−0.331	4.33***
<i>Change in ln(market cap)</i>	0.508	0.159	−0.272	3.25***
<i>Change in ln(assets)</i>	0.142	0.073	−0.113	1.30
<i>Change in ln(sales)</i>	0.121	0.008	−0.089	1.00

Notes. Growth variables are measured from 2002 to 2004. The table shows *t*-statistics for differences in means between treated and control groups and *normalized differences* (defined in Table DD-7 in the online appendix). Significant differences (at 5% level or better) are in bold.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

We provide evidence that the answer is yes in Figure 5. We plot the relation between $\ln(\text{audit fees})$ and 2004 float for Iliev's sample of 281 firms but split them into four groups:

1. Forty-one *shrinker-compliers* (triangles): SOX compliers with float < \$75M in 2004, implying that their float exceeded \$75M in 2002 or 2003.

2. Eighty *grower-compliers* (diamonds): SOX compliers with 2002 float < \$75M but 2004 float > \$75M.

3. One hundred and seventeen control firms (solid circles): firms with 2004 float < \$75M that do not comply with SOX section 404 in 2004.

4. Forty-three *large-compliers* (outlined circles): SOX compliers with float > \$75M in both 2002 and 2004.

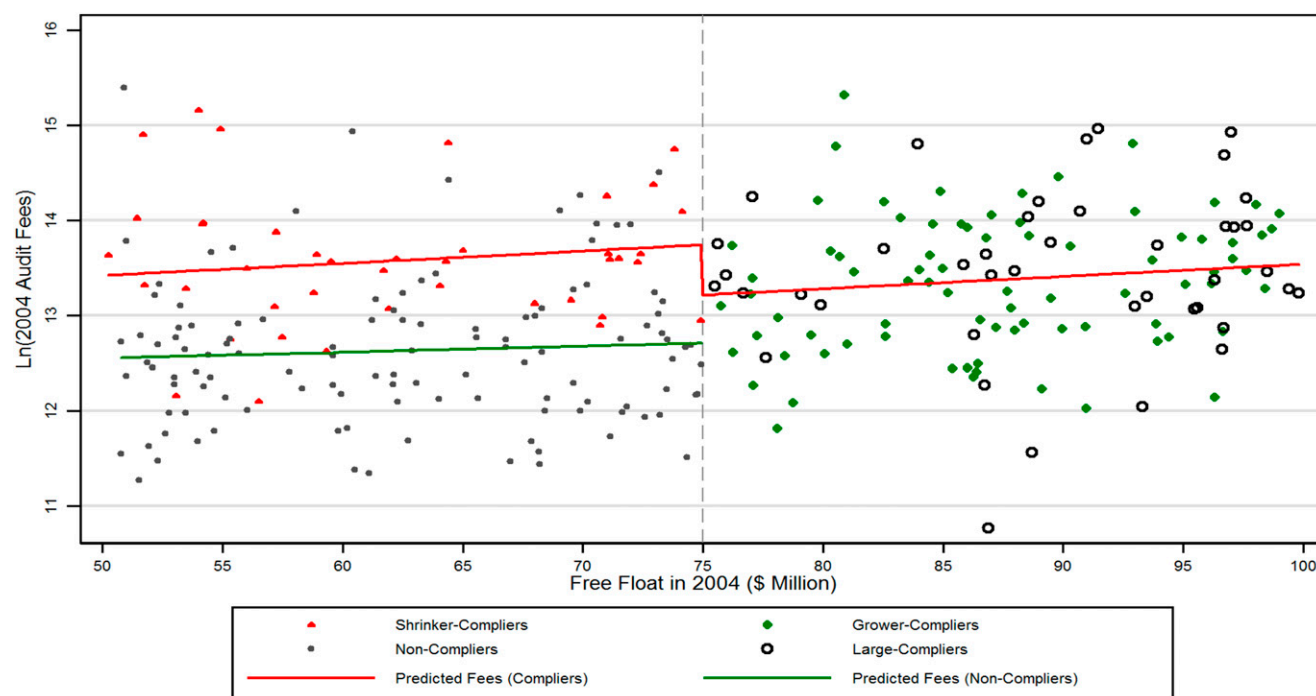
Table 5. Summary Statistics for Covariate Balance in 2004 for 22 Shrinkers (IV Compliers; with 2002 Float > \$75M and 2004 Float < \$75M) and 117 Controls

	Means			
Variable	Controls	SOX compliers	Norm. difference	<i>t</i> -Statistic
Size variables				
<i>ln(float)</i>	4.128	4.125	−0.015	0.09
<i>ln(market cap)</i>	4.385	4.616	0.353	2.18**
<i>ln(sales)</i>	4.221	3.653	−0.138	1.12
<i>ln(assets)</i>	4.467	4.818	0.242	1.36
Iliev's other covariates				
<i>Leverage</i>	0.177	0.090	0.589	0.09
<i>Receivables/assets</i>	0.304	−0.224	−1.295	0.22
<i>Big auditor</i>	0.744	0.298	1.697*	0.30
<i>No. of business segments</i>	2.154	−0.223	−1.333	0.22
<i>No. of geographic segments</i>	1.974	0.153	0.954	0.15
Growth variables				
<i>Change in ln(float)</i>	0.960	−0.566	−0.652	4.19***
<i>Change in ln(market cap)</i>	0.508	−0.582	−0.700	5.76***
<i>Change in ln(assets)</i>	0.142	−0.105	−0.348	2.61**
<i>Change in ln(sales)</i>	0.121	−0.245	−0.144	1.43

Notes. Growth variables are measured from 2002 to 2004. The table shows *t*-statistics for differences in means between treated and control groups and *normalized differences* (defined in Table DD-7 in the online appendix). Significant differences (at 5% level or better) are in bold.

Significance at the 5% level; *significance at the 1% level.

Figure 5. (Color online) Audit Fees vs. Free Float and SOX Section 404 Compliance in 2004



Notes. Natural logarithm of 2004 audit fees versus 2004 free float. The graph shows four groups: (1) 41 shrinker-compliers, which shrink from float > \$75M (in 2002 or 2003) to float < \$75M in 2004 (triangles); (2) 80 grower-compliers, which grow from float < \$75M in 2002 to float > \$75M in 2004 (diamonds); (3) 117 control firms with float < \$75M over 2002–2004 (solid circles); and (4) 43 large-compliers with float > \$75M in 2004 (outlined circles). The full line is from regression for SOX complier firms of $\ln(2004 \text{ audit fees})$ on 2004 float, shrinker-complier dummy (=1 if 2004 float < \$75M), and constant term. Equation is $12.25 + 0.013 \times \text{Float}_{2004} + 0.527 [t = 2.14] \times (\text{shrinker-complier dummy})$. The short regression line ending at float of \$75M is from regression for control firms of $\ln(2004 \text{ fees})$ on 2004 float and constant term. Equation is $12.23 + 0.006 \times \text{Float}_{2004}$. Predicted difference in $\ln(2004 \text{ fees})$ at float = \$75M between shrinker-compliers (other SOX compliers) and control firms = 1.07 (0.54).

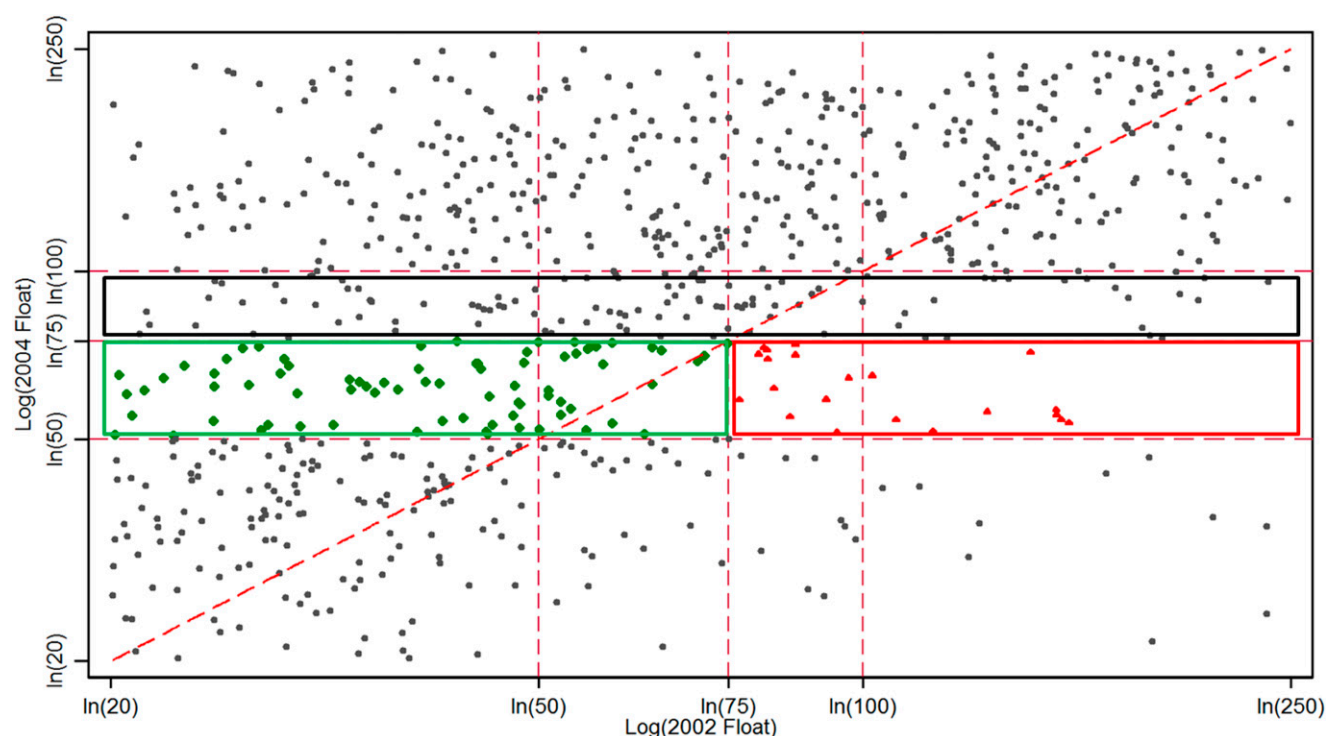
We also add two regression lines showing predicted $\ln(2004 \text{ fees})$. The first line, at the lower left, is for control firms, ends at float of \$75M, and comes from a regression of $\ln(\text{fees})$ on float and constant term. The second line, which covers the full range of 2004 float, is for SOX compliers; we regress $\ln(\text{fees})$ on float, constant term, and a *shrinker-complier dummy* (=1 for shrinker-compliers). This regression line jumps at float of \$75M (higher for lower float), the jump reflects the positive coefficient on shrinker-complier dummy.

The scatter plot and predicted fee lines show that, controlling for float, shrinker-compliers have much higher audit fees than other SOX compliers (the coefficient on the shrinker-complier dummy is 0.53). If we use the regression lines to compare the audit fees of control firms with shrinker-compliers at 2004 float of \$75M, the difference in predicted $\ln(\text{fees})$ is 1.07, similar to Iliev's IV estimate. In contrast, if we compare the same control firms with other SOX compliers with float just above \$75M, the difference in predicted fees is only 0.54—quite close to the DiD/RD estimate that we develop later.

We can now see how not controlling for growth led Iliev to overestimate SOX compliance costs in his IV

analysis. Shrinker-compliers had much higher audit fees than other SOX compliers, without SOX. This is plausible—these firms used to be larger and may have retained the higher fees of larger firms. Some may also have incurred higher fees because they shrank, owing to business troubles, which might call for more intensive auditing. Iliev's IV estimate compares shrinker-compliers with control firms. This produces a large upward bias in estimating the cost of SOX compliance.

Figure 6 illustrates the large differences in growth between IV compliers and control firms. The figure shows a scatter plot of $\ln(2002 \text{ float})$ (x-axis) versus $\ln(2004 \text{ float})$ (y-axis). Dashed vertical lines show 2002 floats of \$50M, \$75M, and \$100M; horizontal lines are similar for 2004. A 45-degree line indicates no change in float from 2002 to 2004. The IV compliers are the 22 shrinkers in the bottom-right bordered box (2002 float > \$75M; 2004 float < \$75M). All are below—often far below—the 45-degree line. The control firms are in the lower-left bordered box (2002 and 2004 floats both < \$75M). Almost all (110 of 117) are above the 45-degree line, often far above it. The largest negative change in $\ln(\text{float})$ among the control

Figure 6. (Color online) Growth Trajectories for Instrument Compliers, Other SOX Compliers, and Control Firms

Notes. Scatter plot of free float in 2002 and 2004 and representation of three groups of firms included in Iliev's study: 22 IV compliers with 2002 float > \$75M but float in 2003 and 2004 float < \$75M shown with triangles in bottom-right bordered box; 119 control firms with 2002 float and 2004 float < \$75M shown with diamonds in bottom-left bordered box; and 142 SOX complier firms with 2002 float > \$75M shown with circles in long upper box. The remaining firms represented with black dots are firms included in the data set provided by Iliev, but fall outside his \$50–\$100M 2004 free float band. We truncate the figure at 2002 and 2004 free float between \$20M and \$250M. Firms above (below) the 45-degree line have an increase (decrease) in float from 2002 to 2004.

firms is -0.20 . The other SOX compliers are in the long upper box, above the \$75M line for 2004.²⁷ We show the imbalance on growth in Table 5. The mean change in $\ln(\text{float})$ from 2002 to 2004 ($\delta \ln(\text{float})$) is -0.57 for IV compliers versus $+0.96$ for controls.

RD's great strength is that it should produce common support and covariate balance on both observed and unobserved covariates, except the running variable (and close substitutes). In this respect, Iliev's IV design grossly departs from his initial RD design. In his creative effort to address one problem (some firms manipulated their 2004 float to avoid complying with SOX), Iliev stumbled into a larger problem with imbalance on growth. That imbalance was pernicious, because shrinkage in float strongly predicts audit fees.

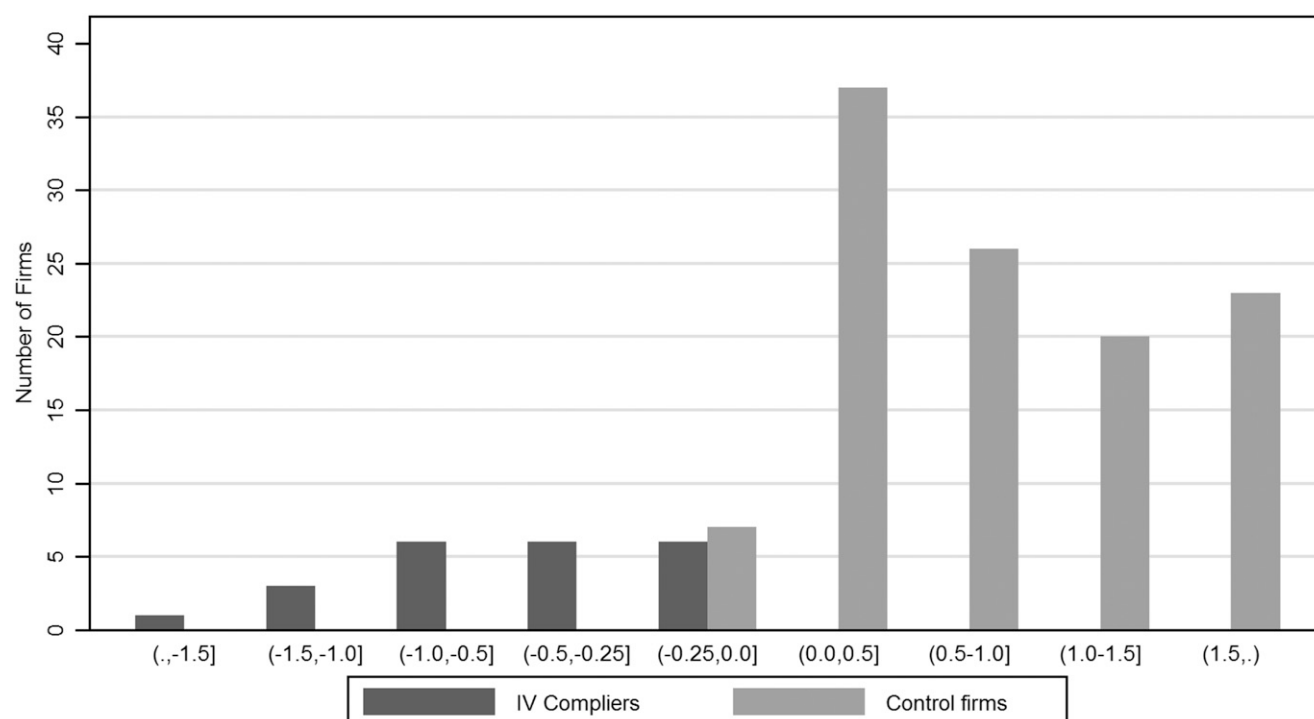
There is very little overlap between IV compliers and controls on growth in $\ln(\text{float})$. Figure 7 shows a histogram for IV compliers and controls within different ranges for $\delta \ln(\text{float})$. There is common support only for seven control firms and six IV compliers with $\delta \ln(\text{float}) \in [-0.25, 0]$. These are the only firms for which IV is credible—manifestly too small a sample to support a reliable estimate. Moreover, the common support sample, even if it were large enough to

support inference, is a narrow slice of all firms, which is unrepresentative of all firms.

In sum, Iliev's IV is undermined by severe imbalance between IV compliers and control firms on growth. His RD design is also affected by imbalance on growth but less severely. We turn below to what else one might do after one recognizes the problem of imbalance on growth. However, we first offer some remarks. Iliev's IV does not satisfy the only-through condition owing to severe imbalance on growth. This imbalance is quite subtle. Iliev was not looking at 2002—he ran cross-sectional regressions in 2004. His RD design should normally produce covariate balance on all but the running variable. He confirmed balance on *levels* in 2004 and likely never suspected imbalance in *changes*. We convinced ourselves, early on, that growth was an important covariate and that his IV estimate was too high. However, for a long time, we did not understand why. After all, controlling for growth reduces the compliance cost estimate only slightly (from 0.744 to 0.706).

We stress that our concern is with the general need for pretreatment balance in shock-based designs (for Iliev, in 2004), not whether particular estimates would survive if researchers required balance. Iliev's

Figure 7. Change in $\ln(\text{Float})$ for IV Compliers vs. Controls



Notes. Histogram of the number of firms within indicated ranges for change in $\ln(\text{float})$ from 2002 to 2004 for 22 IV complier firms (2002 float > \$75M but 2004 float < \$75M) and 116 control firms (float < \$75M over 2002–2004). We drop three of the 119 control firms in the Iliev sample because they report 2002 float > \$75M. There is overlap only in the $[-0.25, 0]$ bin for change in $\ln(\text{float})$. This bin includes seven IV compliers and six control firms.

RD results survive—but they might not have, and his IV results do not.

For us, the takeaway messages from our analysis of Iliev’s IV include:

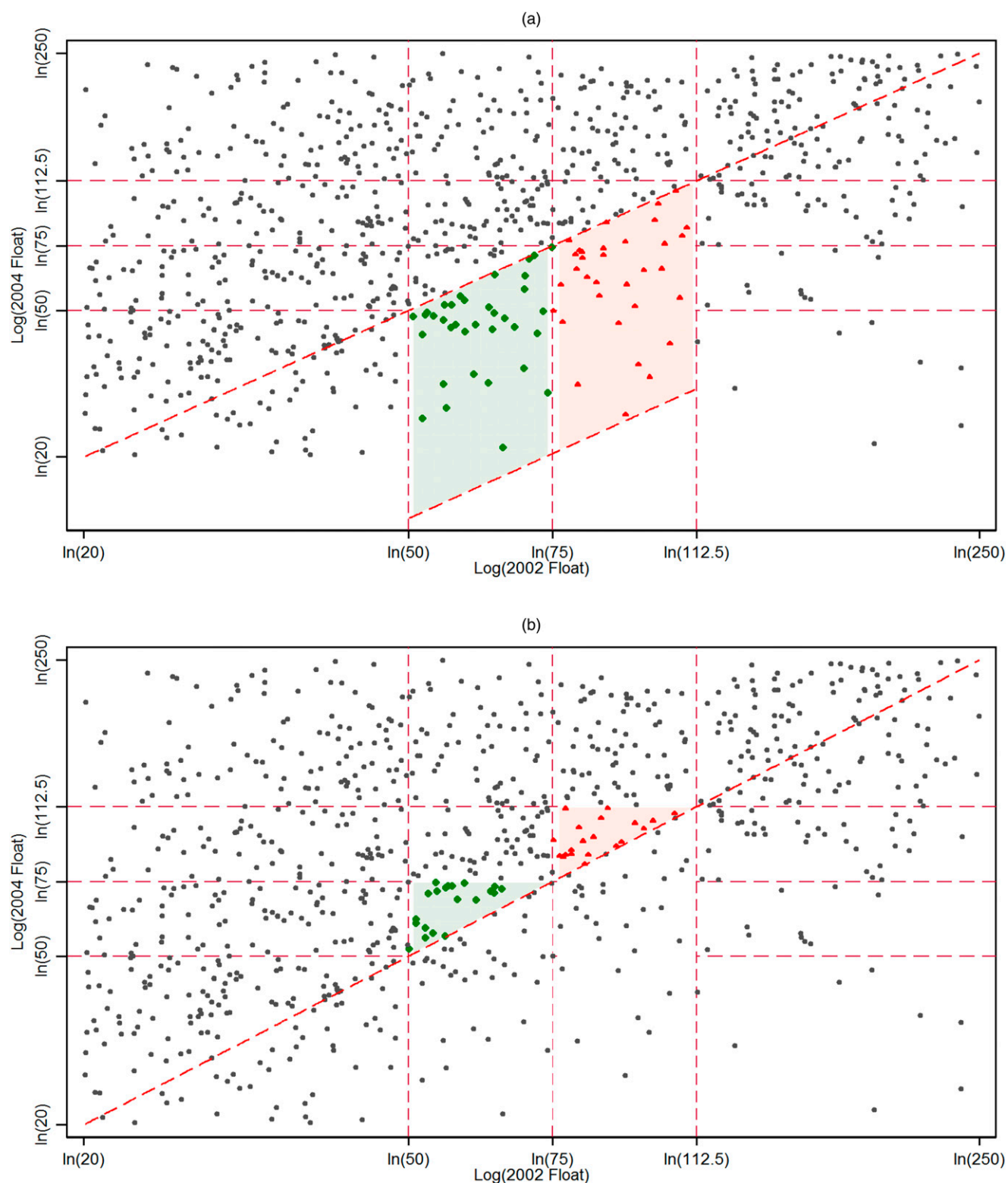
1. Ensuring IV validity can be tricky. An exogenous instrument (which Iliev has) is necessary but not sufficient.
2. One should use extensive control variables and check covariate balance on them. Had Iliev done so, he might have found the growth imbalance and addressed it.
3. A time gap between when the IV is assigned (for Iliev, in 2002) and when it is used (in 2004) can lead to imbalance when the IV is used, which did not exist when it was assigned.
4. Subtle violations of the only-through condition can lead to large biases in IV results.
5. Using regression to extrapolate beyond common support is dangerous.
6. Researchers should discuss whether the IV compliers are likely to be representative of the full sample.

4.5. Our Preferred Analysis: DiD/RD Within Principal Strata Balanced on Growth

We discuss here how one might address imbalance on growth between treated and control firms. We first define a bandwidth in 2002 of 2002 float $\in$ [\$50M,

\$112.5M] (using a bandwidth parameter $b = 1.5$, which we will later vary). We define the bandwidth in 2002 before the shock; Iliev defines his bandwidth in 2004. We address the imbalance on growth between all SOX compliers and all controls by studying subsamples of firms with similar change in float from 2002 to 2004. We define shrinkers as firms with 2002 float > 2004 float. The shrinkers include two principal strata: (1) shrinker-compliers with 2002 float > \$75M that must comply with SOX section 404 regardless of their 2004 float; and (2) shrinker-controls—firms with 2002 float < \$75M. Because compliance is mandatory and no firms comply with SOX section 404 unless required to, both strata are fully observed. Moreover, because firms cannot avoid SOX by shrinking, neither stratum should include SOX avoiders. Iliev’s worry, that SOX avoiders might have relatively high SOX compliance costs, does not apply to the shrinkers.²⁸

We show these two groups in shaded areas in Figure 8(a). Average $\delta \ln(\text{float})$ (the average distance from each point to the 45-degree line) is similar for the two strata; we confirm balance on other covariates, except size covariates, in an unreported covariate balance table. A DiD comparison of $\ln(\text{audit fees})$ for these two groups should not have significant bias owing to imbalance on growth.

Figure 8. (Color online) $\ln(\text{Float})$ in 2004 vs. 2002 for Different Principal Strata

Notes. Scatter plots of free float in 2002 and 2004 and visual representation of treated and control firms included in DiD analysis of shrinkers and modest growers in Table 6. Sample is firms with 2002 float between \$50M and \$112.5M. We truncate the graphs at 2002 and 2004 free float between \$20M and \$250M. (a) Shrinkers strata. Treated firms, represented by triangles, have 2002 float $> \$75\text{M}$; control firms, represented by diamonds, have 2002 float $< \$75\text{M}$. Both groups have 2002 float $>$ 2004 float. Shaded areas represent the additional requirement that $\delta \ln(\text{float}) > -1.3$ ($< 73\%$ drop in float). (b) Modest grower strata. Treated firms, represented by triangles, have 2002 float $> \$75\text{M}$ and 2004 float $< \$112.5\text{M}$. Control firms, represented by diamonds, have 2002 float $< \$75\text{M}$ and 2004 float $< \$75\text{M}$.

Table 6. DiD Analysis Using Principal Strata (Sample with 2002 Float $\in$ [\$50M, \$112.5M])

	Dependent variable: $\ln(\text{audit fees})$								
	Sample: Shrinker strata			Sample: Modest grower strata			Sample: Both strata combined		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Cubic float terms	No	No	Yes	No	No	Yes	No	No	Yes
SOX compliance dummy	0.616*** (3.53)	0.590*** (3.17)	0.606*** (3.25)	0.645*** (3.48)	0.632*** (3.75)	0.507** (2.29)	0.627*** (4.97)	0.578*** (4.43)	0.586*** (4.58)
95% Confidence interval	(0.27, 0.97)	(0.22, 0.96)	(0.23, 0.98)	(0.34, 1.03)	(0.29, 0.97)	(0.06, 0.96)	(0.38, 0.88)	(0.32, 0.84)	(0.33, 0.84)
Implied % increase in fees	85	80	83	91	88	66	87	78	80
Log sales		0.128** (2.49)	0.141** (2.11)		−0.304 (−0.98)	−0.184 (−0.49)		0.105 (1.37)	0.120 (1.46)
Log assets		0.108 (0.61)	0.130 (0.71)		0.996** (2.54)	0.960** (2.41)		0.270* (1.67)	0.285* (1.72)
Log market value of equity		0.023 (0.29)	0.015 (0.18)		−0.078 (−0.79)	−0.090 (−0.89)		0.010 (0.14)	0.001 (0.02)
Leverage		0.162 (0.52)	0.130 (0.45)		−0.450 (−0.93)	−0.342 (−0.70)		−0.038 (−0.14)	−0.056 (0.21)
Receivables/total assets		−0.228 (−0.27)	−0.326 (0.35)		0.981 (1.52)	0.898 (1.28)		0.247 (0.44)	0.177 (0.29)
Big auditor		0.484*** (4.09)	0.465*** (3.51)		0.379*** (2.84)	0.363** (2.64)		0.434*** (4.77)	0.424*** (4.57)
No. of business segments		−0.024 (−0.22)	−0.021 (0.19)		0.024 (0.25)	0.051 (0.51)		−0.021 (−0.22)	−0.018 (0.18)
No. of geographic segments		0.036 (0.47)	0.038 (0.50)		0.096 (0.53)	0.083 (0.42)		0.045 (0.65)	0.048 (0.70)
No. observations	197	197	197	120	120	120	317	317	317
No. firms	67	67	67	40	40	40	107	107	107
No. treated firms	39	39	39	20	20	20	59	59	59

Notes. DiD regressions, with firm and year FEs, of $\ln(\text{audit fees})$ on SOX compliance dummy and indicated covariates using combined DiD/RD design for firms with 2002 float $\in$ [\$50M, \$112.5M]. Sample period is 2002–2004. Sample for shrinkers strata is firms with $\ln(2004 \text{ float}/2002 \text{ float}) \in [-1.3, 0]$ ($<73\%$ drop in float). Treated firms have 2002 float $> \$75\text{M}$; control firms have 2002 float $< \$75\text{M}$. For the modest grower strata (firms with modestly higher float in 2004 than 2002), treated firms have 2002 float $> \$75\text{M}$ and 2004 float $< \$112.5\text{M}$; control firms have 2002 float $> \$50\text{M}$ but $< \$75\text{M}$. Some control firms may have limited 2004 float to avoid SOX compliance. For the both strata combined regressions, we pool the shrinker and modest grower strata. The t -statistics, with standard errors clustered on firm, are in parentheses. Significant results (at 5% or better) are in bold.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

The second pair of strata that we define is for modest growers. These are SOX complier firms with $\delta \ln(\text{float}) > 0$ that have float $> \$75\text{M}$ in 2002 and remain below \$112.5M in 2004 and control firms with 2002 float $< \$75\text{M}$ that remain below the \$75M threshold in 2004. The modest grower stratum and its two substrata of complier and noncomplier firms are shown in triangular shaded areas in Figure 8(b). These two substrata are also balanced on $\delta \ln(\text{float})$ and other covariates, except size and related covariates.

In Table 6, we report the results from DiD regressions with firm and year FEs, comparing $\ln(\text{audit fees})$ of complier and noncomplier firms in the shrinker strata, modest grower strata, and both strata combined. We report three specifications: (1) no covariates other than the fixed effects, (2) adding Iliev's covariates, and (3) also adding a cubic in float. In specification (3), with the strongest covariates, the estimated coefficient on the SOX compliance dummy is 0.606 for shrinkers, 0.507 for modest growers, and 0.586 for

both strata combined (an 80% increase in audit fees). The combined strata estimate is far below Iliev's IV estimate and somewhat below his RD-only estimate of 0.744, although each estimate is within the 95% confidence interval for the other.

4.6. Estimating the Magnitude of Selection Bias

We can also use the principal strata approach to assess the extent of selection bias owing to firms with higher compliance costs being more likely to limit their float to avoid SOX 404 compliance. Selection bias was the main reason for Iliev to use IV. Take the following two “growers” strata: (1) growers-forced-compliers, which are firms with 2004 float $> 2002 \text{ float} > \$75\text{M}$; and (2) growers-voluntary-compliers, which are firms with 2004 float $> \$75\text{M}$ and also 2004 float $> 2002 \text{ float} < \$75\text{M}$ (see Figure Iliev-App-2 in the online appendix). If selection bias is a serious issue, compliance costs should be lower for the second group, because SOX avoiders will have removed

themselves from this group. In fact, in Table Iliev-App-3 in the online appendix, we find only small, statistically insignificant differences between these two strata. The coefficient on the grower-voluntary-complier dummy in regression (3), which has the most extensive covariates, is only -0.037 ($t = 0.48$). Thus, selection bias is limited in practice, and the IV was, therefore, not needed.

4.7. Parallel Trends

For the shrinkers and modest growers strata, we confirm that pretreatment trends are tolerably parallel for the limited pretreatment period for which we have data (2002–2003), although a longer pretreatment period would be desirable. See Figure Iliev-App-3 in the online appendix. We see the expected jump in audit fees for treated firms in 2004, with magnitudes around 0.6, consistent with Table 6.²⁹

5. Reexamination of Desai and Dharmapala (2009)

The third shock-IV paper that we examine is D&D. The authors study the effect of tax avoidance on the value of U.S.-based public firms and whether corporate governance mediates this effect. They find evidence that greater tax sheltering opportunities increase Tobin's q but only for firms with high institutional ownership (which, they posit, are better governed). We discuss D&D briefly in this section, focusing on IV strength. The online appendix provides additional results.

5.1. Research Design

D&D rely for causal inference on a legal rule adopted in late 1996 (*check-the-box rules* for pass-through taxation of nonpublic U.S. firms) as a legal shock. These rules simplified reporting for small private companies. As an unintended by-product, they also reduced tax sheltering costs for large multinational firms. D&D treat the difference between *book* income reported to shareholders and *taxable* income reported to the Internal Revenue Service (book-tax gap, or BTG) as a measure of a firm's tax sheltering activities. They hypothesize that firms with low BTG before the 1996 law change gained more sheltering opportunities than firms with higher BTG. They also hypothesize that at firms with high institutional ownership, greater sheltering will increase firm value (proxied by tax-adjusted Tobin's q), whereas at worse-governed firms, insiders will appropriate the additional value, and therefore Tobin's q will not rise.

BTG, however, is endogenous and can be affected by many firm attributes, some unobserved. D&D address that issue by constructing shock-based instruments

for BTG: they interact the 1996 shock with three *BTG predictors*, which predict the firm's need to shelter its income (net operating loss carryforwards (NOLs), short-term debt, and long-term debt), each scaled by total assets. These variables are also endogenous, but the D&D idea is that interacting them with an exogenous legal shock will make the instruments exogenous as well. The authors address the only-through condition by including these three variables as covariates in their IV analysis. With these controls, it is plausible that the IVs will affect Tobin's q only indirectly through BTG and not directly or through an omitted variable. Thus, D&D, unlike DMO and the other board independence papers, controlled for core endogenous variables that were correlated with their IVs and might generate OVB if omitted.

The D&D research design initially appeared strong. The authors have an exogenous shock and carefully defend exogeneity. The editor and reviewers at *REStat*, a major economics journal, were also convinced. The research design can be questioned. Tax experts told us that BTG could be a poor measure of need for sheltering. Institutional ownership might also be a poor proxy for firm governance. In this paper, we accept the substantive tax and governance theory that underlies their research design and focus on concerns with pretreatment balance, which would undermine their research design even if the substantive theory was right.

D&D report results with firm and year fixed effects and the following covariates: total accruals/assets, option compensation/total compensation, sales, implied volatility of share price, NOLs/assets, short-term debt/assets, long-term debt/assets, |foreign income or loss|/assets, and research and development (R&D) expenditures/assets. Their principal dependent variable is tax-adjusted Tobin's q .³⁰

5.2. Creating a Pre–Post Balanced Data Set

We begin by assessing which firms belong in the sample in a “design” stage of our analysis, with outcomes hidden. The D&D data set includes 862 firms observed at least once over 1993–2001. Of these, 100 are observed only once and effectively drop out with firm FE.

D&D's instruments interact a post-1996 dummy with the endogenous BTG predictors. This design relies on the shock for exogeneity. There is no basis for causal inference for firms that are in the sample only pre-1996 (including 1996) or only post-1996. Yet, of the 762 firms in the effective D&D sample, only 510 appear both before and after the 1996 shock. Of the other 252 firms, 80 appear only pre-1996. Including these firms can affect the coefficients on covariates

but will not directly affect IV estimates because the shock $\times$ covariate IVs are zero in 1996 and earlier. The 172 firms that appear only post-1996 are more troublesome. For these firms, the IVs are identical to the BTG predictors, and are equally endogenous. Thus, one should limit the sample to, at most, the 510 firms that appear both before and after the shock.

Of the remaining 510 firms, 487 have data in both 1996 and 1997. There may be something odd about the 23 firms that pop into and out of the sample in this way. We confirm in unreported regressions that the pop-in firms differ greatly from the other 487 firms on several covariates.³¹ Given these differences and the small loss in sample size, we use a *pre-post balanced sample* of 487 firms with data in both 1996 and 1997 in our principal analyses.

5.3. IV Strength (D&D table 4)

We replicate D&D's firm FE estimates in Table DD-App-1 in the online appendix. In Table 7, we turn to their IV strategy. We replicate D&D's first-stage IV

results for their sample for their two models in regressions (1) and (3) in Table 7 and switch to the pre-post balanced sample in regressions (2) and (4) in Table 7. With the original D&D specification, the instruments are statistically significant but weak, with F -statistics of 3.33 and 3.00, well below the $F > 10$ weak instruments rule of thumb. With the balanced sample, the F -statistics drop to only 1.48 and 1.05; the IVs lose statistical significance and are no longer usable. The greater (though still modest) IV strength with the original D&D sample was driven by post-1996-only firms, for which the instruments are clearly endogenous. We show in the online appendix that their second-stage results and analogous triple-difference results disappear as well (Tables DD-App-2 and DD-App-3).

This paper has emphasized the value of exploiting a shock using both DiD and shock-IV designs. A DiD design would automatically limit the sample to firms that appear both before and after the shock. D&D's error in using a pre-post unbalanced sample would not be possible. Thus, had D&D used both designs,

Table 7. First-Stage IV (D&D table 4) with Original and Pre-Post Balanced Samples

	Dependent variable: BTG			
	Model: Direct effect of shock		Model: Effect mediated by inst. ownership	
	(1)	(2)	(3)	(4)
Sample	Original	Balanced	Original	Balanced
<i>NOL</i> $\times$ <i>post</i>	-0.035** (2.51)	-0.080* (1.74)	-0.016 (0.45)	-0.092 (0.83)
<i>LongTermDebt</i> $\times$ <i>post</i>	-0.013 (0.80)	-0.010 (0.67)	-0.047 (1.23)	-0.045 (1.07)
<i>Current Debt</i> $\times$ <i>post</i>	-0.088* (1.73)	-0.047 (1.27)	-0.310*** (2.59)	-0.241 (1.34)
<i>NOL</i> $\times$ <i>post</i> $\times$ <i>inst. ownership</i>			-0.058 (0.43)	0.037 (0.19)
<i>LongTermDebt</i> $\times$ <i>post</i> $\times$ <i>inst. Ownership</i>			0.055 (0.96)	0.055 (0.87)
<i>CurrentDebt</i> $\times$ <i>post</i> $\times$ <i>inst. Ownership</i>			0.415** (2.52)	0.334 (1.24)
Covariates	Yes	Yes	Yes	Yes
No. of firms	862	487	862	487
No of observations	4,492	3,466	4,492	3,466
Within R^2	0.20	0.26	0.21	0.27
F -test (joint significance of instruments) (p -value)	3.33** (0.02)	1.48 (0.22)	3.00** (0.01)	1.05 (0.39)

Notes. First-stage instrumental variable regressions of BTG (defined in the text) on indicated instrumental variables, with firm and year fixed effects, over 1993–2001. Regressions (1) and (2): Instruments are *NOL*, *Long-Term Debt*, and *Current Debt*, each interacted with *post*dumy. Regressions (3) and (4): Add as additional instruments the interactions between these instruments and institutional ownership. Covariates and sample are same as in Table DD-1 in the online appendix, and variables are defined in Table DD-1 in the online appendix. Noninteracted *post*dummies are absorbed by firm and year effects. Odd-numbered regressions use the D&D sample. Even-numbered regressions use the pre-post balanced sample. The t -statistics clustered on firm are in parentheses. Significant results (at 5% or better) are in bold.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

Table 8. Covariate Balance

Variable	Mean (controls)	Mean (treated)	Norm. difference	<i>t</i> -Statistic
Panel A: Treated = below median BTG in 1996; control = above median				
<i>Tax-adjusted q</i>	2.406	2.311	0.040	0.62
<i>Total Accruals</i>	−0.031	−0.039	0.078	1.23
<i>Options to Total Comp.</i>	0.355	0.352	0.007	0.10
<i>Sales</i>	3.753	3.017	0.076	1.19
<i>Implied Volatility</i>	0.312	0.365	0.252	4.07***
<i>Foreign Losses</i>	0.030	0.034	0.097	1.53
<i>R&D</i>	0.039	0.049	0.120	1.89*
<i>No. Firms</i>	248	239		
Panel B: Treated = above median institutional ownership in 1996; control = below median				
<i>Tax-adjusted q</i>	2.357	2.361	0.002	0.03
<i>Total Accruals</i>	−0.039	−0.032	0.073	1.14
<i>Options to Total Comp.</i>	0.323	0.384	0.173	2.74***
<i>Sales</i>	3.613	3.169	0.046	−0.72
<i>Implied Volatility</i>	0.350	0.326	0.119	1.86*
<i>Foreign Losses</i>	0.030	0.034	0.077	1.21
<i>R&D</i>	0.042	0.045	0.035	0.54
<i>No. Firms</i>	244	243		
Panel C: Treated = below median tax shields in 1996; control = above median				
<i>Tax-adjusted q</i>	2.774	1.954	0.329	5.46***
<i>Total Accruals</i>	−0.032	−0.038	0.054	0.85
<i>Options to Total Comp.</i>	0.368	0.339	0.083	1.29
<i>Sales</i>	2.171	4.587	0.247	3.97***
<i>Implied Volatility</i>	0.366	0.311	0.265	4.28***
<i>Foreign Losses</i>	0.033	0.030	0.063	0.98
<i>R&D</i>	0.054	0.033	0.250	4.03***
<i>No. Firms</i>	241	246		

Notes. Summary statistics on covariate balance for the pre-post balanced sample of 487 firms (3,466 observations) with data in both 1996 and 1997. Dollar amounts are in millions. The table shows means for covariates in 1996 for three ways to define treated and control firms. In Panel A, firms are treated if they have below-median BTG in 1996. In Panel B, firms are treated if they have above median institutional ownership in 1996. In Panel C, firms are treated if they have below-median tax shields in 1996. Covariates are defined in Table 7. The table shows the *t*-test for differences in covariates x_j (indexed by j), $|t_j| = |\bar{x}_{jt} - \bar{x}_{jc}| / [(s_{jt}^2/N_t + s_{jc}^2/N_c)]^{1/2}$, where s_{jt} and s_{jc} are the standard deviations for treated and control groups, respectively, and N_t and N_c are the numbers of treated and control firms, respectively. The table also shows absolute values of *normalized differences* (Imbens and Rubin, 2015) defined as $ND_j = |\bar{x}_{jt} - \bar{x}_{jc}| / [(s_{jt}^2 + s_{jc}^2)/2]^{1/2}$. Significant differences at 5% level or better are in bold.

*Significance at the 10% level; ***significance at the 1% level.

they would have realized that something was wrong with their IV design.

5.4. Covariate Balance and Parallel Trends

For an IV based on an interaction, one should confirm covariate balance in several ways that probe various aspects of the interaction. We illustrate this in Table 8. We present results on covariate balance for the pre-post balanced sample using three divisions of firms into treated and control based on 1996 attributes. In panel A of Table 8, we divide firms into treated and control based on the instrumented variable BTG. Treated firms have below-median BTG and hence, in the D&D story, higher need for tax sheltering; control firms have above-median BTG. In panel B of Table 8, we divide the sample based on institutional

ownership in 1996. Treated firms have above-median institutional ownership and hence, in the D&D story, should benefit more from tax sheltering opportunities. In panel C of Table 8, we divide firms based on a measure of need to shelter taxable income using the covariates that enter the D&D instruments. To construct this measure, we rank each firm based on the three instruments (NOL, short-term debt, and long-term debt) and use the sum of the three ranks to define a dummy variable $highTaxShieldNeed96 = 1$ for treated firms with below-median sum of ranks and 0 otherwise.³²

In each panel, we report means for the treated and control groups for 1996 for the D&D covariates. We also report *t*-statistics for a two-sample test for difference in means and normalized differences.

In panels A and B of Table 8, treated and control firms are relatively balanced on most covariates. However, low-BTG firms have significantly more share price volatility, and firms with high institutional ownership grant more options as a share of total executive compensation. Balance is substantially worse in panel C of Table 8 when we classify firms based on the covariates that D&D use to predict need to shelter income proxied by *highTaxShieldNeed96*.

These differences between treated and control firms increase the risk of nonparallel trends during the treatment period. If the D&D results had survived to this stage, one would want to address this imbalance using balancing methods. We illustrated this process in our reexamination of DMO. In fact, the D&D sample exhibits nonparallel trends on their outcome, Tobin's q , both before the shock and well after it (long enough after the shock so that this is unlikely to be a delayed response to treatment; see the online appendix, Figure DD-App-1). Mean Tobin's q rises for the control firms relative to treated firms in 1996, just prior to treatment; it rises again in 1999, well after the shock, and then it falls in 2000 and 2001. The changes over 1999–2001 suggest differing reactions to the technology bubble of the late 1990s, which deflated in 2000 and 2001. Given these nonparallel changes at times unrelated to the shock, even if one found an effect on Tobin's q for (say) 1997 versus 1996, one could not ascribe it with confidence to the shock.³³

6. Improving Shock-IV Research Design Through Pretreatment Balance

In this part, we provide a checklist of steps that researchers should take to improve and assess pretreatment balance in shock-IV designs drawing on our earlier reexaminations, explain selected items, and assess how the papers in our sample fare on the checklist items. Most items also apply to DiD designs. The online appendix provides an extended checklist, including additional items not directly related to pretreatment balance.

6.1. A Shock-IV Checklist Focusing on Pretreatment Balance

There are only two classical IV requirements for instrument validity: instrument strength and an untestable *exclusion restriction* $\text{Cov}(z, \epsilon) = 0$. Causal IV theory separates the exclusion restriction into separate exogeneity and only-through conditions. Our reexamination highlights the need to defend each of these conditions. It also highlights the need for pretreatment balance as a precondition for plausibly as if random assignment to treatment. No research design checklist can cover every situation, but our analysis suggests the following checklist for shock-IV

studies intended to ensure that a research design satisfies the conditions for a good shock, especially pretreatment balance.

1. *Assess and carefully defend exogeneity.* Why should the shock be expected to create plausibly as if random assignment to treatment, either directly or conditioned on covariates?

2. *Check for common support.* Check for common support on potentially relevant pretreatment covariates and pretreatment outcomes. Common support can often be ensured through sample trimming. Severe lack of overlap may require abandoning the project.

3. *Check covariate balance.* Check preshock balance on potentially relevant pretreatment covariates and pretreatment outcomes, and report results in a covariate balance table. For an IV design with an interaction term, one needs balance on both components of the interaction. Moderate imbalance can often be improved using balancing methods. Severe imbalance may require abandoning the project.

4. *Use extensive covariates to assess common support and improve pretreatment balance.* If the shock is truly clean, using extensive, potentially relevant covariates should not greatly affect estimates. If the design is less clean, extensive covariates can limit the sources of violation of the exogeneity and only-through conditions. Some covariates may reduce IV strength; this normally indicates that they are needed and is not a justification for omitting them.

5. *Assess and carefully defend as if random assignment to treatment.* One needs, at a minimum, plausibly as if random assignment conditioned on the covariates. An ideal instrument will produce similar treated and control groups without conditioning on covariates.

6. *Assess pretreatment parallel trends.* Use DiD methods to assess whether parallel trends are present during the pretreatment period. Data permitting, do so for a long pretreatment period.³⁴ Use graphical methods, such as a leads-and-lags graph. If the research design uses an interaction variable, check for parallel trends for both components of the interaction. For mild nonparallelism, look for additional covariates that can reduce the extent of the nonparallel trends. Severe nonparallelism may require abandoning the project.

7. *Assess and carefully defend the only-through condition.* Ensuring pretreatment balance can help in defending this condition.

8. *Confirm instrument strength after trimming and balancing.* Many shock-IVs will weaken after trimming and balancing. Present first-stage IV results after these steps. Insufficient strength will often require abandoning the project.

9. *Present OLS, shock-IV, and intent-to-treat DiD estimates.* Present, for the same sample, OLS, intent-to-treat DiD, and IV estimates (both stages). Assess

whether the IV coefficient magnitudes are economically plausible.

10. *Assess IV blowup*. If the IV coefficient is much larger than the OLS coefficient, defend both the economic plausibility of the IV coefficient and why a larger IV coefficient is plausible given the expected sign of the expected OLS bias.

11. *Graph your results*. Leads-and-lags graphs are a start, but often, more can be done. Our reexaminations of DMO and Iliev suggest some possibilities.

12. *Run placebo tests*. The details will vary based on the design, but placebo tests will often be available. One can vary the time of the shock; test whether the shock predicts backward in time; if the shock relies on a threshold, vary the threshold; and test placebo outcomes.

6.2. Defending Exogeneity

All shock-IV designs assume that the shock is exogenous and (perhaps conditioned on covariates) as if randomly assigned. Exogeneity cannot be directly tested. Researchers must rely instead on logic. They need to argue that a shock is exogenous for the specific firms in their sample based on theory or institutional knowledge. D&D and Iliev provide excellent examples of how to do so. DMO are on more slippery ground—before the Audit Committee Shock, firms chose whether to have nonfully independent audit committees. This design weakness is not fatal, but it puts stress on ensuring pretreatment balance.³⁵

Exogeneity also has falsifiable implications. A truly exogenous shock should generate as if random assignment and thus both common support and covariate balance as well as parallel pretreatment trends. Lack of common support can be addressed through trimming; imbalance on the trimmed sample, if not too severe, can be addressed through balancing methods. If, after these steps, one has nonparallel pretreatment trends, there is often no convincing solution.

6.3. Assessing and Enforcing Common Support

One theme of this paper is that ensuring reasonably thick common support should be a central element of shock-IV designs in all shock-based designs. Yet, no paper in our sample assesses common support.

The concept of reasonably thick support has no precise definition but can be illustrated by example. For DMO, we trimmed to $PctIndep_{2000} \in (0.25, 0.80]$. Our reasoning was as follows. There are many treated firms but only three control firms with $PctIndep_{2000} \leq 25\%$; we would not want treatment effects to depend on possibly idiosyncratic choices by a few control firms. For $PctIndep_{2000} > 80\%$, one could include the six treated firms in the sample, but the logic behind *Audit Committee IV*—that firms need to add independent

directors to their boards to staff a fully independent audit committee—is absent for these firms.³⁶

For DMO, we assessed and enforced common support on a core covariate. More commonly, one assesses common support on the propensity score and does not insist on strict common support for each covariate that enters the propensity score. Instead, two firms with similar propensity scores are considered similar enough to be fairly compared. Usually, one assesses common support and seeks to balance on the same set of covariates.

6.4. Assessing and Improving Covariate Balance

Trimming the sample to common support will usually improve balance, but often, more will be needed. A second core theme of this paper is the importance of assessing balance on pretreatment covariates and outcomes and using balancing methods to address significant imbalance. A few recent DiD papers use matching or other balancing strategies, but that practice has not percolated into IV studies. Five of the 26 panel data papers in our sample provide a covariate balance table. Of these, Tsoutsoura (2015) has balance. Balakrishnan et al. (2014) uses matching to improve balance, and it is the only paper in our sample to do so. Iliev (2010) has balance as of 2002 but imbalance on growth as of 2004. The remaining two papers (DMO, Armstrong et al. 2014) lack balance but do not to address the imbalance they find.

It is beyond our scope to discuss the available balancing methods and how to choose among them.³⁷ We note, however, that many balancing approaches are not compatible with two-stage estimation, and their properties for two-stage estimation are not known. One can use a broader set of methods for single-stage estimation, including assessing whether the first stage remains usably strong after trimming and balancing.

6.5. Which Pretreatment Covariates Should Not Be Balanced On?

When a shock does not provide good balance on pretreatment covariates and outcomes (after trimming), we have advocated balancing on a broad range of potentially relevant pretreatment covariates, which plausibly predict assignment to treatment. We address here the other side of the coin—which covariates should one not balance on? We discuss relevant considerations later. All require judgment.

First, our core advice is to balance on relevant *covariates*—variables that meaningfully predict the treatment or the outcome. No one proposes balancing on irrelevant variables. Theory can provide guidance on which variables are relevant. One can also assess, in the design stage of a project, which variables predict assignment to treatment. At the same time, apart from the specific exceptions discussed later,

balancing on additional covariates usually does little harm. If one balances on $2n$ covariates (say) instead of n , balance will often slightly worsen for the first n but only slightly in our experience. One hopes that by balancing on a broad set of observable covariates, balance on unobserved covariates also improves, and it is the unobserved covariates that generate potential OVB.³⁸

The principal exception to the advice to balance on a broad set of relevant covariates, not relevant for our replications, is that one should not balance on a variable that is likely to be a “collider” in Judea Pearl’s terminology (Pearl 2009; Morgan and Winship 2014, chapter 3)—a variable that is caused by the IV but does not cause the IV. The concept of a collider variable generalizes the standard advice not to condition on an intermediate outcome of a treatment; most real-world colliders involve intermediate outcomes (Elwert and Winship 2014).

Second, if firms can anticipate treatment and change a covariate before treatment occurs, this covariate is an intermediate outcome, and balancing on it can introduce bias. One response with panel data is to balance farther back in time prior to the shock before anticipation is likely. One can use a leads-and-lags graph for the suspect covariate to assess the evidence for anticipation.

Third, one should be cautious in balancing on a “rare” binary covariate with probability close to zero (or one), which does not strongly predict treatment. These covariates are hard to match on. Matching algorithms seek improved balance across all covariates and may try too hard to achieve balance on the rare covariate at the cost of worse balance on more important covariates.³⁹

6.6. Balancing on Pretreatment Covariates vs. Regression on Time-Varying Covariates

Balancing on pretreatment covariates, both fixed and time varying, is reasonably safe. Regression on using time-varying covariates is much less so. Suppose that we can divide the time-varying covariates into *affected* covariates (potentially affected by treatment) and *unaffected* covariates. Including unaffected covariates can increase precision and will not cause bias. Including affected covariates can lead to bias. Yet, it is often unclear which covariates fall into which group.

A further consideration is that including time-varying covariates in a regression analysis can reduce the likelihood and extent of nonparallel trends. If so, one must trade off reducing bias from nonparallel trends with inducing bias by controlling for affected covariates. This calls for judgment informed by theory plus robustness checks for the effects of including (versus not) *suspect* covariates. A leads-and-lags graph for the suspect covariate can help in assessing whether it seems to be affected by the treatment.⁴⁰

6.7. Assessing Parallel Trends

A careful check for parallel trends should be an obligatory part of shock-based design. Assessing placebo shocks can be valuable but will usually be subsumed by a leads-and-lags graph for a reasonably long pretreatment period. Moreover, a placebo shock during the pretreatment period cannot substitute for a good graph. A trend that is visible in a graph can coexist with an insignificant outcome for one or even several placebo shock dates. Four of the 26 six-panel shock-IV papers in our sample assess whether pretreatment trends are parallel, but none provides a true leads-and-lags graph.⁴¹ We provide an example of such a graph in Figure App-1 in the online appendix.

Researchers sometimes address nonparallel trends by adding linear time trends to a regression specification that also includes firm and year FEs. However, linear trends will rescue the design only if any nonparallelism is linear and if the linear trend would have continued in the posttreatment period but for the shock. Yet, a trend without known cause might stop or even reverse in the postshock period. Balancing the treated and control groups can sometimes reduce the nonparallel pretreatment trends; one can also search for, and add covariates that absorb the nonparallelism.

6.8. Defending the Only-Through Condition

All IV designs assume that the IV affects the outcome only through the instrumented variable. Like exogeneity, this assumption cannot be directly tested and must be defended through logic and a design that seeks to exclude other channels. D&D and Iliev provide good examples of how to do so.⁴² An exogenous shock is not enough to ensure that the only-through condition is satisfied. However, confirming balance for a rich set of covariates can help. For example, Iliev might have caught the subtle violation of only through for his design if he had assessed balance on growth.

In some studies, including DMO and Iliev, the only-through condition will be plausible only if one ensures balance on key covariates. A linear or polynomial control for the key covariates might suffice. However, a more robust approach is to divide the sample into blocks (or principal strata) on a key covariate or on propensity to be treated, estimate treatment effects within blocks, and then sum across blocks. We illustrate this approach for Iliev in the text.

6.9. Instrument Strength

Enforcing common support and covariate balance often reduces instrument strength, as our three reexaminations illustrate. There is no solution for this: shocks that satisfy as if random assignment and are useably strong are hard to find. One should assess and report whether the IV remains useably strong after trimming and balancing.⁴³

A reasonable first-stage t -statistic (or an F -statistic for two or more instruments) is necessary but not always sufficient. It can also be useful to assess statistical power—what is the likelihood that the overall design can detect effects of plausible size? If that likelihood is low, the project often cannot be rescued. Black et al. (2019) provide an example where limited first-stage power and limited second-stage power (how strongly does the instrumented variable plausibly predict the outcome) combine to produce insufficient overall power.⁴⁴

6.10. Comparing OLS and IV Coefficients and IV Coefficient Blowup

Pretreatment balance can be seen as necessary but not sufficient for a research design to plausibly satisfy the only-through condition. Another plausibility check is to compute the ratio of (IV coefficient/OLS coefficient). A high ratio is a warning for possible violation of the only-through condition; the higher the ratio, the stronger the warning. Sometimes, as in Iliev, even a ratio modestly above one can be a warning sign if the IV coefficient is economically implausible. We believe that all IV papers should report both OLS and IV coefficients. We want to see four estimates—regress x on z (2SLS first stage), regress y on x (OLS), regress y on z (intent to treat), and the 2SLS estimate—plus discussion of the differences.

Of the 16 papers in our sample that report both OLS and IV results from comparable specifications, only four papers have a ratio below 1.5 (including Iliev at 1.3), whereas seven have troubling ratios above five, including D&D at 5.8 and DMO at 7.5 for share returns (Figure App-2 in the online appendix). We concur with Jiang (2017) that a high IV/OLS coefficient ratio should call for—but in our sample never receives—careful defense of why the IV coefficient is economically plausible and why one would expect the OLS coefficient to be strongly biased toward zero. Defense will sometimes be possible. The true causal effect of x on y could be masked in OLS by reverse causation or omitted variable bias.⁴⁵ Researchers should also address whether the compliers are likely to be representative of all firms.

A high IV/OLS ratio can also be defended by ensuring pretreatment balance. We expect that, in many cases, lack of pretreatment balance drives a high IV/OLS ratio. If authors ensure pretreatment balance and they still have an acceptably strong IV, the blowup problem may lessen. We cannot confirm this intuition for the papers that we reexamine, however, because of their first stages when we impose pretreatment balance.

6.11. Time Gap Between Shock and IV Use

A risk for shock-IV design, illustrated by Iliev, involves situations when the shock occurs before the instrumented variable is measured. Such a time gap exists for 6 of the 23 papers in our sample.⁴⁶ The structure of Iliev's IV leads to very different growth for IV compliers and controls between the shock (2002) and its use (in 2004), which violates the only-through condition. If a time gap exists, researchers need to assess covariate balance both at the time of the shock and at the later time when it is used and whether the time gap might introduce OVB not present at the time of the shock. None of the time gap papers do so.⁴⁷

7. Concluding Remarks: On Shock-IV and the Causal Inference Enterprise

The increased emphasis on causal inference in corporate governance and accounting research has led to increased reliance on IV, shocks (using a variety of research designs), and sometimes, shock-IVs. Larcker and Rusticus (2010) explain why most nonshock-IVs used in corporate finance and accounting research are flawed; Jiang (2017) shows that many IVs exhibit coefficient blowup. This paper provides evidence that apparently good shock-IVs often cannot support valid causal inference, because they fail to satisfy the need for pretreatment balance, including common support, covariate balance, and pretreatment parallel trends. A realm for valid shock-IV exists, but it seems to be narrow. How narrow is unknown, because the steps in our checklist are often not taken.

Perhaps the best that we can do, as researchers, is work to improve our causal inference methods while remaining skeptical about new methods. For example, most early DiD papers did not cluster standard errors. Bertrand et al. (2004) explained why this is problematic, and Petersen (2009) showed that alternatives to clustering do not work for panel data. The apparent statistical significance of many results in early papers using panel data without clustered errors is likely spurious. There is unavoidable fragility in new methods. The need for skepticism is deepened by concern about whether results can be replicated (Chang and Li 2015). We did not face that problem for the core results in DMO, Iliev, and D&D.

We hope that our reanalysis will be understood as an effort to improve causal methods. Researchers who are considering a shock-IV design should ensure that the shock satisfies the conditions for a good shock, including the need for pretreatment balance, and that IV is a sensible use of the shock.

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Endnotes

¹ In contrast, the authors of two other papers using IVs similar to DMO refused to share data or even to tell us which firms were in their samples so that we could ourselves obtain the (public) data that they rely on.

² Distance-to-Compliance IV was first used by Armstrong et al. (2014); it equals 0 for firms with a majority-independent board in 2000 and $(50\% - \text{PctIndep}_{2000})$ for other firms.

³ See Armstrong et al. (2014) (using Distance-to-Compliance IV), Chen et al. (2015) (using Audit Committee and Minority-Independent IVs), and McMartin and Needles (2016) (using Distance-to-Compliance IV). Similar DiD papers include Chhaochharia and Grinstein (2007) (DiD design using a composite measure of pre-SOX compliance based on four measures, two of which are fully independent audit committee and majority-independent board), Barger et al. (2010) (using triple-difference design with SOX Board Shock as the basis for the third difference), Andrade et al. (2014) (triple-difference design using the Chhaochharia and Grinstein measure), Chen (2014) (DiD design using SOX Board Shock), Dah et al. (2014) (DiD design assuming that SOX Board Shock caused firms with majority-independent boards to reduce board independence), Bradley and Chen (2015) (DiD design using SOX Board Shock), Donelson et al. (2016) (DiD design using SOX Board Shock), and Kim and Klein (2017) (DiD design using Audit Committee Shock).

⁴ See Ahern and Dittmar (2012), Dale-Olsen et al. (2013), Matsa and Miller (2013), Bertrand et al. (2014), Bohren and Staubo (2014), and Eckbo et al. (2016).

⁵ The SEC has used the \$75M threshold in other rules; Iliev and other authors have exploited this threshold in a series of papers. See Gao et al. (2009), Chhaochharia et al. (2011), Kinney and Shephardson (2011), Lee et al. (2014), Weber and Yang (2019), Dharmapala (2016), Ge et al. (2017), and Gustafson and Iliev (2017).

⁶ The need for as if random assignment to treatment goes under various names in the causal inference literature, including the “conditional independence assumption” (Angrist and Pischke 2009), “unconfounded” or “ignorable” treatment assignment (Imbens and Rubin 2015), and “selection on observables” (Imbens and Wooldridge 2009). A shock-IV must also satisfy monotonicity—being assigned to

treatment cannot cause firms to reject a treatment that they would otherwise have chosen (Angrist et al. 1996).

⁷ Judea Pearl’s directed acyclic graphs may be helpful in understanding this point (Pearl 2009, Morgan and Winship 2014). In Pearl’s graphical approach to causal inference, one must close all *back-door paths* to obtain valid inference, *especially* strong ones. For DMO, one such path is Audit Committee Independence $\leftarrow$ Board Independence $\rightarrow$ Firm Outcomes. Yet, Duchin et al. (2015) advocate leaving this path open *because* it is strong.

⁸ Compare Angrist and Pischke (2009) with Imbens and Rubin (2015). Both discuss the differences between matching and regression; Angrist and Pischke (2009) are more willing to accept regression as a substitute for matching.

⁹ The generalization of Abadie’s *local average response function* to a continuous IV, called the *marginal treatment effect*, comes from Heckman and Vytlacil (2005) and Heckman et al. (2006); see French and Taber (2011) for a readable explanation.

¹⁰ We report sample sizes with $\ln(q)$ as the outcome.

¹¹ There can be exceptions. Suppose that balance (after trimming) is reasonably good, coefficient estimates are similar with and without balancing, and statistical significance is found without but not after balancing. Researchers could disclose this and argue that they have “some” or “mild” support for the hypothesis, that a treatment effect exists. However, they cannot do so without conducting the balancing exercise. Compare Ho et al. (2007) (advocating matching prior to regression to reduce model dependence and researcher discretion to present only selected strong results).

¹² We restrict the sample for this comparison with firms used in the Tobin’s q regressions (with nonmissing values for all covariates and δQ). We impose the same restriction in reporting first-stage results in Table 2 and in creating Figures 1–4. The usable sample varies slightly depending on the outcome variable.

¹³ DMO in their reply (Duchin et al. 2015, p. 4) defend not controlling for PctIndep_{2000} because it is closely “intertwined” with audit committee independence. It is precisely because these variables are closely related that one *must* control for PctIndep_{2000} to separate the exogenous effect of audit committee IV on board independence from endogenous changes in board independence that the firm would have made for other reasons.

¹⁴ We explain this choice of bounds further in Section 3.6.

¹⁵ The correlation is -0.64 for the 323 treated firms ($t = 15.1$) and -0.59 for the 582 control firms ($t = 17.8$).

¹⁶ Other examples of shocks based on a board percentage threshold include (a) minimum requirements for women on boards of directors, including Norway’s requirement for 40% women, and (b) other board independence shocks, such as the Cadbury Committee recommendation for at least three independent directors on UK boards (studied by Dahya and McConnell 2007). Compare studies of how firm leverage affects other firm outcomes, which have long struggled with whether firms seek to maintain a target leverage ratio, to which they gradually return following a shock to leverage (see the survey by Graham and Leary (2011)).

¹⁷ See section 6.3 for further discussion of this trimming choice. There are no hard rules on how to trim to reasonably thick common support. However, one should choose the trimming bounds in a design stage of the overall research plan before studying outcomes. In unreported robustness checks, the results that we present below are not sensitive to the choice of trimming bounds.

¹⁸ First, DMO instrument for δIndep using a first-stage regression of δIndep on noncomply dummy and covariates. They then interact instrumented δIndep (the predicted value of δIndep from the first-stage regression) with InfoCost ; this interaction is their main variable of interest in the second stage. One instead needs to use separate instruments for δIndep and $\delta \text{Indep} \times \text{InfoCost}$: the natural instruments

are noncomply dummy and noncomply dummy $\times$ *InfoCost*. Second, DMO do not include *InfoCost* as a covariate in their main results, but any regression with an interaction term should include the non-interacted components.

¹⁹ Good DiD practice includes using panel data from the pretreatment period to assess whether pretreatment trends are parallel for the outcome variable. Pretreatment trends should also be parallel for the covariates. Here, we propose assessing parallel trends for *PctIndep*, a covariate that strongly predicts the instrumented variable δIndep .

²⁰ This predictive power disappears, as it should, if we address imbalance on *PctIndep* in the placebo shock year. See Table DMO-App-4C in the online appendix.

²¹ In Figure DMO-App-1 in the online appendix, we show that the *Minority Independent IV* also produces identical V-shaped graphs and that reflect regression to the mean.

²² Of these, 91 control firms are used once, 43 firms are used twice, 8 are used three times, 8 are used four times, 3 are used five times, 2 are used six times, and 1 firm is used seven times. Thus, the results are not driven by using one or a few firms, perhaps unrepresentative, as matches.

²³ To construct Figures 4(A2) and 4(B2), we use the same trimming and matching strategy except applied in the pseudoshock year. Thus, for a pseudoshock in 1998 (1999), we trim and match in 1998 (1999).

²⁴ The standard errors reported in the first- and second-stage regressions do not account for the sample matching and trimming. Because the propensity score is estimated, these errors are biased. This bias is typically upward, is usually small, and can be addressed using bootstrapping methods. See Wooldridge (2010), chapter 21.

²⁵ Within a panel data design, it can be valuable to use both FE and first-differences designs. This lets one assess whether the results from the two approaches are similar; if they are not, this casts doubt on the usual exogeneity-of-covariates assumptions that apply to both designs (Grieser and Hadlock 2019).

²⁶ Iliev's main results use a bandwidth of [\$50M, \$100M] in 2004 float. He finds similar (unreported) results with bandwidths of [\$60M, \$90M] and [\$40M, \$110M].

²⁷ Per Iliev's data on float, three control firms had 2002 float $>$ \$75M. We drop these firms in our reanalysis because Iliev's data on reported 2002 float must not be the float that the SEC used to determine whether SOX section 404 compliance was required.

²⁸ In unreported results, we confirm covariate balance between the shrinker-compliers and shrinker-controls on all variables except size-related variables, including balance on growth variables.

²⁹ In Figure Iliev-App-3(c) in the online appendix, we also check parallel trends for the third pair of strata: grower-voluntary-compliers (seen as treated firms) versus growers-forced-to-comply. There is some nonparallelism from 2002 to 2003 (which drives the insignificant negative coefficient in Table Iliev-App-3 in the online appendix) but no evidence of a pseudo-treatment effect from 2003 to 2004. This provides further evidence that selection bias for the SOX avoiders is not an important issue in practice.

³⁰ D&D use individual firm dummies; we also do so for consistency. One could also use demeaned values (similar to panel regressions with fixed effects in non-IV settings) using the *xtivreg2*.ado module for Stata; the standard errors will be slightly different.

³¹ For each covariate, we regress the covariate on a pop-in dummy (=1 for these 23 firms and 0 otherwise), year dummies (omitting 1996), and constant term. The constant term then gives the mean for non-pop-in firms in 1996. The coefficients on the pop-in dummy are often large relative to the constant term. For NOLs, the coefficient on the constant term is near zero at 0.011; the coefficient on pop-in dummy is 20 times larger at 0.231, although statistically insignificant. Pop-in firms also have about half the sales and twice the R&D/assets as other firms.

³² An alternative way to assess balance with a continuous IV is to regress the IV (as dependent variable) on the covariates. The coefficients on the covariates should be small and insignificant. See French and Song (2014).

³³ Compare with Catan and Klausner (2017), who discuss problems that can arise in a DiD setting if treated and control firms are not balanced on a pretreatment covariate—for them, Tobin's q .

³⁴ Our experience is that pretreatment trends that appear reasonably parallel over a limited pretreatment period often appear less so over a longer period.

³⁵ Only seven of the remaining shock-IV papers in our sample defend exogeneity: Dharmapala et al. (2011), Hendershott et al. (2011), Cornelli et al. (2013), Bach (2014), Balakrishnan et al. (2014), and Tsoutsoura (2015).

³⁶ Good practice, which we followed, is to choose trimming bounds without looking at outcomes, but it turns out (Figure 1) that all six of these firms reduced their board independence between 2000 and 2005.

³⁷ See generally Imbens and Rubin (2015). For trimming to common support, see Crump et al. (2009); for matching, see Rosenbaum (2009). For inverse propensity weighting, see Busso et al. (2014). If a shock-based IV is binary, a balance table is easy to construct. For a continuous IV, one can instead regress each covariate on the IV and assess whether the IV predicts the covariates (French and Song 2014).

³⁸ This is a hope, with no guarantee. Balancing on one endogenous covariate can sometimes worsen balance on another unobserved covariate. For example, in the DMO setting, balancing on *PctIndep*₂₀₀₀ could worsen balance on the proportion of independent directors on committees other than the audit committee (not observed), because the control firms have more independent directors on the audit committee and hence, perhaps fewer on other committees. Not balancing on the observed covariate will rarely improve matters, yet balancing on it will be imperfect as well.

³⁹ An alternative is to remove the observations with unusual values for the rare binary covariate. As an example, suppose that a sample of people is 98% female. One might decide to remove the men from the sample and present estimates for women rather than balancing on gender.

⁴⁰ Our advice differs from the standard advice in the causal inference literature, which is to control only for covariates that are clearly not affected by the treatment. This is the right advice for cross-sectional data (the focus of most of this literature) and also if assignment to treatment is unconfounded, which this literature assumes.

⁴¹ DMO, Dharmapala et al. (2011), Dam and Koetter (2012), and Bach (2014). However, even when parallel trends are assessed in some way, the assessment is often limited. For example, DMO provide a graph of *industry-adjusted* ROA (ROA is one of their three outcomes) for control firms relative to treated firms split into three groups based on *InfoCost*. They do not provide graphs for *PctIndep*, non-industry-adjusted ROA, or other outcomes and covariates. Fracassi and Tate (2012), Bach (2014), and Balakrishnan et al. (2014) use placebo shocks during the pretreatment period.

⁴² Four other papers in our sample assess the only-through condition: Cornelli et al. (2013), Bach (2014), Balakrishnan et al. (2014), and Tsoutsoura (2015).

⁴³ Of the 11 shock-IV papers with a single instrument (including Iliev), nine report first-stage results. Of these, eight have a t -statistic $>$ 3.3 and thus satisfy the $F > 10$ rule of thumb for avoiding the classical weak instruments problem. Of the 15 multiinstrument papers (including D&D and DMO), 11 report first-stage results. Of these, 7 have $F > 10$. However, this is before addressing common support and covariate balance.

⁴⁴ Randomization inference can provide correct confidence intervals even with a weak instrument (Imbens and Rosenbaum 2005), but it is never used in our sample.

⁴⁵ A classic example comes from criminal justice. If cities respond to crime by hiring police, then in OLS, more police will predict more crime. If police reduce crime, a valid IV for police hiring should have the opposite coefficient. See Chalfin and McCrary (2018).

⁴⁶ The papers are Cohen et al. (2012), Fracassi and Tate (2012), Goetz et al. (2013), Balakrishnan et al. (2014), Cole et al. (2014), and Tsoutsoura (2015). Extreme examples of a time gap between IV assignment and assessment of outcomes include the legal origin instruments popularized by La Porta et al. (1998) and the settler mortality instrument in Acemoglu et al. (2001).

⁴⁷ For the gender of first-born child IV used by Tsoutsoura (2015) and earlier used by Bennedsen et al. (2007) (not a panel data paper), Roberts and Whited (2013) observe that families with first-born girls may transfer control of a family firm to their daughters only when the daughters are especially talented. This provides another example of how a time gap between when an instrument is assigned (first-born birth) and when the instrument is used to predict the instrumented variable (founder succession, often decades later) can induce imbalance between treated and controls.

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